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Master Thesis project

MASTER'S THESIS

"Evaluation of baseline wander removal approaches in ECG recordings"

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Master Thesis project

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Abstract

Cardiovascular Diseases (CVD) rank first among the causes mortality of the population of our planet according to the World health organization (WHO), accounting for about a third of all cases of death. In Europe, almost a million people die from these diseases every year (at least 50% of all deaths), of which about 100 thousand are people at working age. In addition, experts (WHO) forecast a further increase in cardiovascular morbidity and mortality, as in developed and developing countries, due to aging populations and lifestyle features. According to preliminary data, in 2030 myocardial infraction and stroke will remain the main individual causes of death [1].

In this regard, there is a need for accurate and timely diagnosis of diseases of the cardiovascular system in the early stages development, which is one of the strategic directions of the health care industry. The basis of such a diagnosis is electrocardiography (ECG). This method is a non-invasive examination of the bioelectrical activity of the heart. Currently this method of diagnosis is indispensable for rhythm and conduction disorders, hypertrophy of the ventricles and atria, coronary heart disease, including painless form, myocardial infarction and other dangerous pathologies.

However, in connection with the peculiarities of physiological origin and registration of signals, there is a serious and unsolved problem of their automatic analysis. ECG signals are unsteadily structured signals with cyclically repeating informative sections. Specifically, according to these signs, the state of the cardiovascular system is assessed in electrocardiography. Morphology and parameters are diverse, variable, and unpredictable.

In addition, it should be borne in mind that, while electrocardiography is recorded, it is inevitable in record also numerous interferences and noises in the signal. There are various kinds and origins that influence signal, leading to morphological changes in the electrocardiogram. In some cases, such changes do not have diagnostic importance, but sometimes they can be mistakenly considered as signs of myocardial damage, leading to inaccurate or erroneous automatic conclusions.

As a result of this problem, in cardiac diagnostics is development of various methods and algorithms for noise suppression. Have been developed each method for such problems has advantages and disadvantages, therefore, why in this project we analyze different methods to struggle against interferences. The effectiveness of noise suppression will be determined by the degree of suppression of interference and the degree of preservation of diagnostic information.

Improving the efficiency of noise suppression makes it possible to increase the reliability of automatic conclusions, and ultimately contributes to improving the quality of prevention and treatment.

The aim of this work is to develop and test methods for suppressing interference. The main objectives of the thesis are the following:

1) research about general information about electrocardiography;

2) review types of interferences which are arising in electrocardiograms;

3) overview methods and algorithms for interference suppression in electrocardiograms;

4) develop methods which overcome baseline wander interference;

5) test the designed filters;

6) compare filters by using different methods for suppressing interference.

Project overview

1. Objectives

The main objective of this project is to design filters which can detect baseline wander noise from ECG signal and get rid of it. In this work we will implement and compare several methods which are widely used for the elimination of baseline wander.

2. Structure

Introduction overviews existing problems with cardiovascular diseases nowadays and indicates ways to improve the quality of prevention and treatment of these diseases by showing the importance of finding new methods of struggling against interferences in electrocardiogram.

Chapter 1 puts biomedical signal processing in context and gives a brief description of bioelectricity and its manifestation on the body surface as signals. General aspects on signal acquisition and performance evaluation are briefly considered.

Chapter 2 contains a background to the electrophysiology of the heart and describes the main characteristics of the ECG signal in terms of morphology and rhythm.

Chapter 3 contains an overview of the main types of interference arising during registration. The reasons for their occurrence are also considered. Then it describes a suite of methods, essential to any system which performs ECG signal analysis, developed for the purpose of noise reduction, heartbeat detection and delineation, and data compression.

Chapter 4 Overview such methods of filtering as Linear, Time-Invariant Filtering and Forward-backward IIR filtering.

3. Description

Cardiovascular diseases are the leading cause of death all over the world. As a result, there is a growing demand for diagnostic and therapeutic tools necessary for the study and treatment of patients with these abnormalities. The electrocardiogram (ECG) is the simplest and most common non-invasive technique in the diagnosis of heart diseases. The ECG is strongly affected by some types of noise, like baseline wander. Baseline wandering is a typical problem in ECG records, that generally produces artefactual data while measuring ECG parameters. Baseline wander is a low frequency artifact in the ECG that arises from breathing, electrically charged electrodes, or subject movement, and can hinder the detection of changes in the ST segment (which is essential for diagnosing some cardiac pathologies) because of the varying electrical isoline. Hereford, even minor fluctuations in baseline can lead to the decision if a patient is classified as ST-Elevation Myocardial Infarction (STEMI) or non-STEMI

and thus influence the therapeutic approach dramatically. Baseline interferences are especially relevant in ECG records obtained during exercise testing. The frequency components of the baseline wander are usually below 0.5 Hz but, in case of stress tests, this limit can be higher. Thus, these components can be in the same range than the low frequency ECG components, like those of the ST segment. Then, removal of the baseline can adversely alter the ECG clinical information.

Several methods have been used to eliminate the baseline wander. The most classical one is **ensemble averaging**. This approach is adequate when the ECG signal remains constant in each beat, but this is not the situation in many actual ECG records. Other method is **polynomial interpolation**. Linear interpolation introduces significant distortions. A third order approximation called cubic spline achieves better results. Interpolation makes use of a previous knowledge of the ECG isoelectric levels estimated from the PR intervals, also called knots. This is a nonlinear approach whose performance depends on the knots determination accuracy, and it is degraded as the knots become more separated (low heart rate). Another method that overcomes this problem is digital **highpass linear phase filtering**. This method can be implemented in real time, but has two major problems: first, the filter requires either to be an FIR filter with a long impulse response and a large number of coefficients or an IIR filter with forward-backward filtering (thus precluding on-line implementation); second, given that ECG and baseline wander spectra usually overlap, it is not possible to remove the baseline wander with a linear filter without distorting the ECG components. The **time-varying filtering** technique, that selects different cut-off frequencies of the linear filter as a function of the heart rate or the baseline level, improves the time invariant FIR filter performance, but can yet distort the ST components and has high computational requirements. Adaptive filtering has been recently proposed to cancel the baseline drift. This filter is an adaptive transversal filter with one weight, where the reference input is a constant with a value of 1 and the primary input is the ECG signal. This filter, using the LMS algorithm in the adaptation process, is equivalent to a linear highpass filter, that takes the advantage of the adaptive implementation, but still modifies the ST components.

4. Methodology

In this work we will implement several methods which are used in the literature for the elimination of baseline wander in the ECG, comparing their performance in terms of the attained mean squared error (MSE): starting from a simple high pass fitter, and then using cascade adaptive filters to preserve the overlapped deterministic low frequency components of the ECG, such as ST segment components. This cascade adaptive filter works in two stages. The first stage is an adaptive notch filter at zero frequency. The second stage is an adaptive impulse correlated filter that estimates the ECG signal correlated with the QRS occurrence. In both stages the LMS algorithm is used with different adaptation factors $\mu 1$ and $\mu 2$. We will analyse the frequency response of the filter as a function of the $\mu 1$ and $\mu 2$ parameters, selecting those more appropriated for baseline removal. Finally, the performance of the filter is studied on an actual ECG affected by baseline drift and we will compare the results after using different types of filters with each other and with clean ECG signals (like signal 118m from the MIT-BIH database).

5. Tasks and timetable of fulfillment

The main tasks performed in the thesis is here been following:

- Download the clean ECG signals and the base line wander noise from Physionet.
- Design a high pass filter with optimal parameters.
- Create cascade adaptive filter preserving the overlapped deterministic low frequency components of the ECG
- Compare results of implemented algorithms.

Table 1.1 Shows the timetable for different tasks.

Table 1.1 – Timetable of fulfillment assigned tasks

Week/ Task	1	2	3	4	5	6	7	8
Defining goals								
Literature study								
Designing filter's								
Implement filters								
Recording test results								
Presentation								

6. Materials to be used

The following in materials have been used in this project:

- Software: Windows 10 Enterprise x64 (1703 build 15063), MATLAB R2019b.
- Hardware: Apple MacBook Pro 13 A1502.

Chapter 1

Biomedical Signal Processing

1.1. Introduction

Chapter 1 puts biomedical signal processing in context and gives a brief description of bioelectricity and its manifestation on the body surface as signals. General aspects on signal acquisition and performance evaluation are briefly considered. The chapter consist of:

1.1 – introduction.
1.2 – show the main objectives of biomedical signal processing.
1.3 – represents the purpose of major clinical contexts for biomedical signal processing.
1.4 – fundamentals in bioelectrical signals.
1.5 – discussion – show short overview of chapter 1.

1.2. Objectives

A fundamental objective of biomedical signal processing is to **reduce the subjectivity** of manual measurements [2]. The introduction of computer-based methods for the purpose of objectively quantifying different signal characteristics is the result of a desire to improve measurement accuracy as well as reproducibility. In addition to reducing measurement subjectivity, biomedical signal processing is used for developing methods that extract features to help characterize and understand the information contained in a signal. Such feature extraction methods can be designed to mimic manual measurements, but are equally often designed to extract information which is not readily available from the signal through visual assessment. For example, small variations in heart rate that cannot be perceived by the human eye have been found to contain very valuable clinical information when quantified in detail using a suitable signal processing technique. Although it is certainly desirable to extract features that have an intuitive meaning to the physician, it is not necessarily those features which yield the best performance in clinical terms [4].

In many situations, the recorded signal is corrupted by different types of noise and interference, sometimes originating from another physiological process of the body. For example, situations may arise when ocular activity interferes with the desired brain signal,

when electrodes are poorly attached to the body surface, or when an external source, such as the sinusoidal 50/60 Hz powerline, interferes with the signal. Hence, **noise reduction** represents a crucial objective of biomedical signal processing so as to mitigate the technical deficiencies of a recording, as well as to separate the desired physiological process from interfering processes. In fact, the desired signal is in certain situations so dramatically masked by noise that its very presence can only be revealed once appropriate signal processing has been applied. This is particularly evident for certain types of transient, very low-amplitude activity, such as evoked potentials, which are part of brain signals, and late potentials, which are part of heart signals.

Certain diagnostic procedures require that a signal be recorded on a long timescale, sometimes lasting for several days. Such recordings are, for example, routinely achieved for the purpose of analyzing abnormal sleep patterns or to identify intermittently occurring disturbances in the heart rhythm. The resulting recording, which often involves many channels, amounts to huge data sizes, which quickly fill up hard disk storage space once a number of patients have been examined. Transmission of biomedical signals across public telephone powerlines is another increasingly important application in which large amounts of data are involved. For both of these situations, **data compression** of the digitized signal is essential and, consequently, another objective of biomedical signal processing. General-purpose methods of data compression, such as those used for sending documents over the Internet, do not perform particularly well, since the inherent characteristics of the biomedical signal are not at all exploited. Better performance can be obtained by applying tailored algorithms for data compression of biomedical signals[3]. Data compression can also be understood in a wider sense as the process in which clinical information from a long-term recording is condensed into a smaller data set that is more manageable for the person analyzing the data. In this latter sense, it is highly desirable to develop signal processing algorithms which are able to determine and delimit clinically significant episodes.

Mathematical signal modeling and simulation constitute an other important objectives in biomedical signal processing which can help to attain a better understanding of physiological processes. With suitably defined model equations it is possible to simulate signals which resemble those recorded on the cellular level or on the body surface, thereby offering insight into the relationship between the model parameters and the characteristics of the observed signal. Examples of bioelectrical models include models of the head and brain for localizing sources of neural activity and models of the thorax and the heart for simulating different cardiac rhythms. Signal modeling is also central to the branch of signal processing called "model-based signal processing," where algorithm development is based on the optimization of an appropriately selected performance criterion [7]. In employing the model-based approach, the suggested signal model is fitted to the observed signal by selecting those values of the model parameters which optimize the performance criterion. While model-based biomedical signal processing represents a systematic approach to the design of algorithms, it does not always lead to superior performance; heuristic approaches may actually perform just as well and sometimes even better. It is a well-known fact that many commercial, medical devices rely on the implementation of “ad hoc” techniques in order to achieve satisfactory performance [2].

The complexity of a signal model depends on the problem to be solved. In most signal processing contexts, it is fortunately not necessary to develop a multilevel model which accounts for cellular mechanisms, current propagation in tissue, and other biological properties. Rather, it is often sufficient to develop a "phenomenological" model which only accounts for phenomena which are relevant to the specific problem at hand [4].

1.3. Clinical context

The purpose of this section is to point out the three major clinical contexts in which algorithms for biomedical signal processing are designed [4], namely, the contexts of

- Diagnosis.
- Therapy.
- monitoring.

In the **diagnostic context**, medical conditions are identified from the examination of signal information, reflecting the function of an organ such as the brain or the heart, in combination with other symptoms and clinical signs. A signal is often acquired by a noninvasive procedure, which makes the examination less taxing on the patient. Most of these procedures are also associated with inexpensive technology for acquisition and analysis, thus increasing the likelihood that the technology can be disseminated to countries with less developed economies. A diagnostic decision rarely requires immediate availability of the results from signal analysis, but it is usually acceptable to wait for some time(from a few minutes to some hours or even days) for the analysis to be completed. Hence, signal analysis can be done off-line on a personal computer, thus relying on standardized hardware and operating system, possibly supplemented with a digital signal processor (DSP) board for accelerating certain bottleneck computations. Algorithms for biomedical signal processing do not define the entire diagnostic computer system, but their scope ranges from performing a simple filtering operation to forming a more substantial part of the clinical decision-making.

Therapy generally signifies the treatment of disease and often involves drug therapy or surgery. With regard to biomedical signal processing, therapy may imply a narrower outlook in the sense that an algorithm is used to directly modify the behavior of a certain physiological process, for example, as the algorithms of a pacemaker do with respect to cardiac activity. In a therapeutic context, an algorithm is commonly designed for implementation in an implantable device like a heart defibrillator, and, therefore, it must, unlike an algorithm operating in a diagnostic context, strictly comply with the demands of on-line, real-time analysis. Such demands pose some serious constraints on algorithmic complexity as well as on the maximal acceptable time delay before a suitable action needs to be taken. Low power consumption is another critical factor to be considered in connection with devices that are implanted through a surgical procedure; for example, the battery of an implantable device is expected to last up to ten years. Hence, algorithms which involve computationally demanding signal processing techniques are less suitable for use in a therapeutic context.

Biomedical signal processing algorithms form an important part of real time systems for **monitoring** of patients who suffer from a life-threatening condition. Such systems are usually designed to detect changes in cardiac or neurological function and to predict the outcome of a patient admitted to the intensive care unit (ICU). Since such changes may be reversible with early intervention, irreversible damage can sometimes be prevented. Similar to therapeutic contexts, the signal is processed during monitoring in an essentially sequential fashion such that past samples constitute the main basis for a decision, while just a few seconds of the future samples may also be considered, which usually stands in sharp contrast to signal processing for diagnostic purposes, where the signal is acquired in its entirety prior to analysis. Thus, a noncausal approach to signal analysis can only be adopted in the diagnostic context, which mimics that of a human reader who interprets a signal by making use of both past and future properties. Constraints need to be imposed on the algorithmic design in terms of maximal delay time, because the occurrence of a life-threatening event must be notified to the ICU staff within a few seconds. Other important issues to be considered are the implications of a clinical event that is missed by the algorithm or the implications of a nonevent that is falsely detected causing the staff to be notified.

1.4 Basics of Bioelectrical Signals

Bioelectrical signals are related to ionic processes which arise as a result of electrochemical activity of a special group of cells having the property of **excitability**. The mechanisms which govern the activity of such cells are similar, regardless of whether the cells are part of the brain, the heart, or the muscles. In particular, the electrical force of attraction has central importance for the processing and transmission of information in the nervous system, as well as for sustaining the mechanical work done by the heart and the muscles.

The ability of excitable cell membranes to generate **action potentials** causes a current to flow in the tissue that surrounds the cells. With the tissue being a conducting medium, commonly referred to as a volume conductor, the collective electrical activity of many cells can be measured noninvasively on the body surface. The recording of a bioelectrical signal in clinical practice is done by attaching at least two electrodes to the body surface. The "exploring" electrode, placed close to the electrical source, and the "indifferent" electrode, placed elsewhere on the body surface. Multiple electrode configurations are commonly used in clinical practice to obtain a spatial description of the bioelectrical phenomenon. Since the activity of excitable cells is viewed from a distance by the electrodes, with different tissues in between, such as blood, skeletal muscles, fat, and bone, it is impossible to noninvasively determine detailed information about cellular properties and propagation patterns. Nonetheless, significant empirical knowledge has been acquired over the years from analyzing the patterns of signals recorded on the body surface, which have been found crucial for clinical decision-making.

The problem of characterizing the electrical source by noninvasive measurements has, in spite of the above-mentioned limitations, been the subject of considerable research due to

the far-reaching clinical implications of its potential solution. In order to arrive at a meaningful solution, it is necessary to introduce a mathematical model in which the collective electrical cellular activity is treated as a volume source, i.e., it is defined by a fixed dipole, a multiple dipole, or some other source model. Furthermore, by introducing a model for the volume conductor which accounts for essential properties of the human body, such as geometry and resistivity, the electrical field measured on the body surface can be modeled. The important inverse problem consists of determining the electrical source from measurements on the body surface under the assumption that the geometry and electrical properties of the volume conductor are known.

The properties of signals which take place call for widely different processing techniques; an individual waveform can in some signals be directly linked to a specific clinical diagnosis, while in other signals many waveforms must be analyzed before a meaningful interpretation can be made. This project deals with the processing of electrical signals that describe the activity of the heart.

The electrocardiogram (ECG) reflects the electrical activity of the heart and is obtained by placing electrodes on the chest, arms, and legs, **see Figure 1.1**. With every heartbeat, an impulse travels through the heart, controlling its rhythm and rate, and causing the heart muscle to contract and pump blood. The ECG represents a standard clinical procedure for the investigation of heart diseases such as myocardial infarction.

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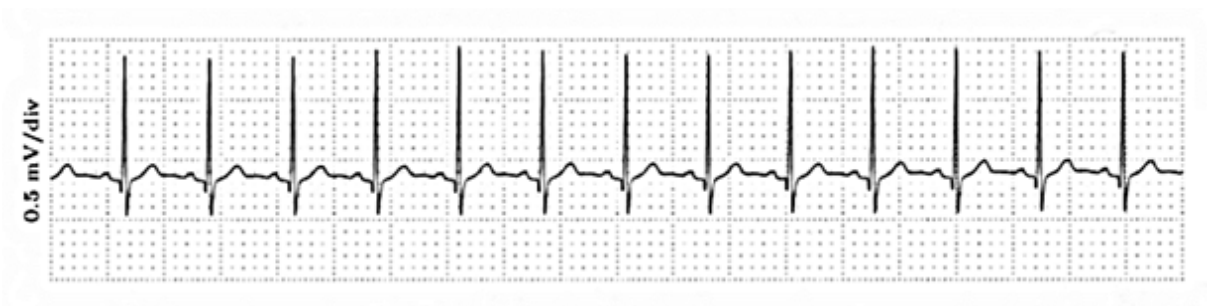

Figure 1.1 Example of bioelectrical signal - an electrocardiogram (ECG), recorded from the body surface [5].

1.5 Discussion

The main objectives of biomedical signal processing are the following:

- **reducing the subjectivity** of manual measurements by applying computer-based methods for the purpose of objectively quantifying different signal characteristics. As a result - improving measurement accuracy and reproducibility.
- **noise reduction -** mitigating technical deficiencies of a recording and separating the desired physiological process from interfering processes. Clean signals are

always masked by noise and interference and they can only be revealed once appropriate signal processing has been applied.

- **data compression**. Struggles with easily filled disk storage by, for example, long timescale signal recordings, which in consequence would be more complicated for transmission. Also, in clinical information from a long-term recording is condensed into a smaller data set which is more manageable for the person who analyzing the data. Better performance can be obtained by applying tailored algorithms for data compression of biomedical signals.
- **Mathematical signal modeling** and simulation could help to attain a better understanding of physiological processes. With suitably defined model equations it is possible to simulate signals which resemble those recorded on the cellular level or on the body surface, thereby offering insight into the relationship between the model parameters and the characteristics of the observed signal.

Clinical contexts for biomedical signal processing are:

- **Diagnosis** - examination of signal information, reflecting the function of an organ such as the brain or the heart, in combination with other symptoms and clinical signs.
- **Therapy** - treatment of disease and often involves drug therapy or surgery. With regard to biomedical signal processing, therapy may imply a narrower outlook in the sense that an algorithm is used to directly modify the behavior of a certain physiological process.
- **Monitoring** - real time systems for observing patients who suffer from a life-threatening condition, which may be reversible with early intervention and irreversible damage could be prevented.

The electrocardiogram (ECG) reflects the electrical activity of the heart and is obtained by placing electrodes on the chest, arms, and legs. With every heartbeat, an impulse travels through the heart, controlling its rhythm and rate, and causing the heart muscle to contract and pump blood. The ECG represents a standard clinical procedure for the investigation of heart diseases, such as myocardial infarction.

Chapter 2

Concepts of electrocardiography

2.1 Introduction

Chapter 2 contains a background to the electrophysiology of the heart and describes the main characteristics of the ECG signal in terms of morphology and rhythm. The chapter consist of:

2.1 – introduction.

2.2 – provides information about ECG: how it is generated, its main components and parameters. In pathology, these parameters can vary significantly, which makes it possible to use electrocardiography in the diagnosis of many heart diseases.

2.3 – demonstrate the role of signal processing in analyzing ECG recordings.

2.4 – shows introduction in types of filtering ECG signals – linear and nonlinear.

2.5 – discussion – show short overview of chapter.

2.2 Structure, characteristics and parameters of electrocardiograms

Electrocardiography - electrophysiological study of heart activity based on registration and analysis of the electrical activity of the myocardium, which leads to the pervasive heartbeat. The curve obtained as a result of the process of electrocardiography is called an electrocardiogram.

This method of studying the bioelectrical activity of the heart is today indispensable in the diagnosis of rhythm and conduction disorders, hypertrophy of the ventricles and atria, coronary heart disease, myocardial infarction and other heart diseases [8].

The main advantages of electrocardiography are [2] :

- high information content and reliability.
- painlessness and safety.
- efficiency of research.
- lack of contraindications.

An electrocardiogram is a signal which is displaying the contractile activity of the heart and can be easily recorded using surface electrodes placed on the limbs or chest. The electrocardiogram, isprobably, the most widely known, recognized, and used biomedical signal.

2.2.1 Electrical phenomena in the heart

The occurrence of electrical phenomena in the heart is based on a change in the concentration of potassium (K^+), sodium (Na^+) and calcium (Ca^{2+}) ions, as well as chlorine (Cl), inside and outside myocardial cells under the influence of an electrical impulse. In moment of rest, the outer surface of the membrane is positively charged, and the inner is negatively charged due to the uneven distribution of charged ions. This potential difference between the intracellular and extracellular environment is called the **resting potential (RP)** and is approximately 80 mV in the myocardium of the atria and ventricles, 90 mV in the His-Purkinje system(see Figure 2.2), and 60 mV in the sinus (SU) and atrioventricular nodes (AVN) [4].

The concentration of Na^+ ions is 20 times higher in the extracellular medium than inside the cell, Cl^- ions is 13 times higher and Ca^{2+} ions is 25 times higher. On the contrary, the concentration of K^+ ions inside the cell is 30 times higher than in the extracellular fluid. Such high ion concentration gradients on both sides of the membrane are maintained due to the functioning of ion pumps in it, with the help of which Na^+, Ca^{2+} and Cl^- ions are removed from the cell, and K^+ ions enter the cell. If an electric stimulus of sufficient strength is applied to the cell membrane to change the resting potential value to the level of the **threshold potential**, rapid **depolarization** occurs - a sudden change in charge inside the cell to positive (up to +20 or +30 mV). Depolarization, or zero phase, of the **action potential (PD)** is due to the rapid entry of sodium ions into the cell.

After depolarization (phase 0) a much slower process of **repolarization** begins - the restoration of the initial resting potential. In the cells of the working myocardium and His-Purkinje system, the first phase of repolarization (phase 1) occurs rather quickly and is due to the release of K^+. It is believed that in phase 1, the current Cl^- takes part inward. In the plateau phase (phase 2), against the background of the current K^+, Ca^{2+} (and to a lesser extent Na^+) enters through the so-called — slow channels. As a result, the speed of depolarizing and repolarizing currents is balanced for a time and an action potential plateau appears. At the end of the plateau, the slow channels begin to close, and the conductivity for K^+, on the contrary, increases sharply — the polarization accelerates (phase 3) and a return to the initial resting potential level occurs. After this, diastole begins (phase 4).

In the cells of the CA-site and the AB-compound, fast sodium channels are absent. Therefore, the depolarization of the membranes of these cells is almost completely determined by the slow incoming current of Ca^{2+}. Since the intensity of this current is small, and its duration reaches 5–10 ms, the phase 0 of PD nodal cells has a relatively small slope. Figure 2.1 [6] shows the action potentials of the sinoatrial node (a) and working cardiomyocytes (b).

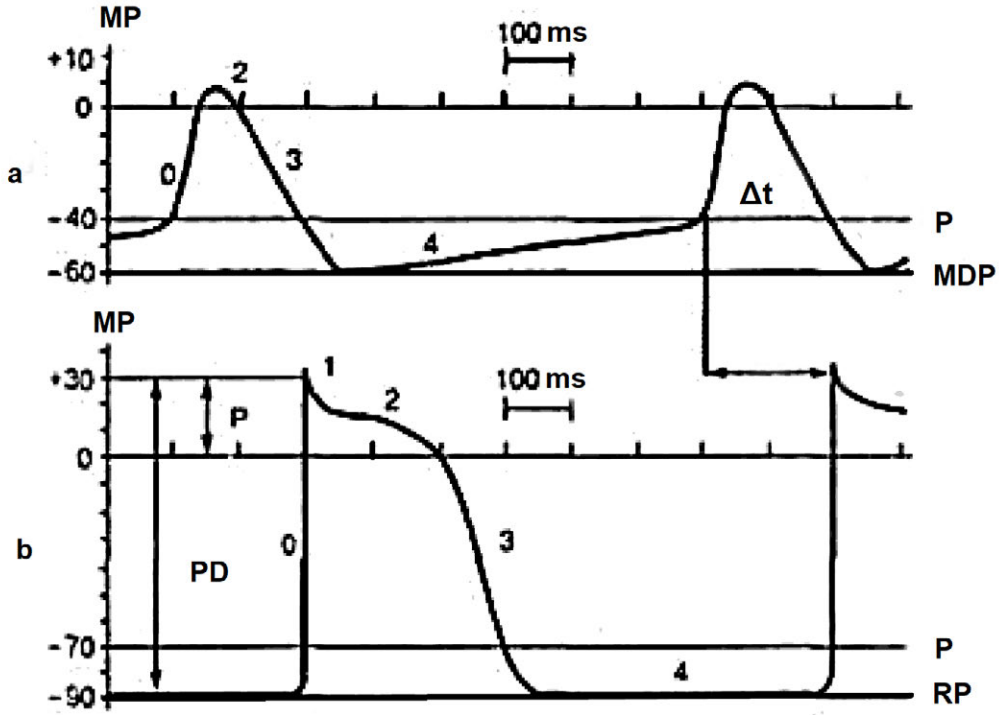

Figure 2.1 Potential action of cardiomyocytes

a - the sinoatrial node; b - working cardiomyocytes [6]

The numbers in Figure 2.1 denote: 0 - the phase of rapid depolarization, 1 - the phase of initial repolarization, 2 - the phase of slow repolarization (plateau), 3 - the phase of rapid final repolarization, 4 - the phase of spontaneous diastolic repolarization.

The letters indicate: MP (mV) - membrane potential, P - threshold potential (critical level of depolarization), MDP - maximum diastolic potential, RP (TMPP) - resting potential, PD (TMPD) - amplitude of the action potential, P - reversal (overshoot) membrane potential, Δt - is the time of excitation from the CA site to the ventricles.

All these processes are related to the excitation of a single muscle fiber of the myocardium. Having arisen during depolarization, the impulse causes excitation of neighboring sections of the myocardium, which gradually covers the entire myocardium and develops as a chain reaction (Figure 2.2). Myocardial electrical excitation begins in a special zone of the right atrium - in the sinus node, which is the pacemaker, as it contains pacemaker cells. In a healthy person, the sinus node generates electrical impulses with a frequency of 60-90 beats per minute, which then propagate to the atria along the atrial conduction paths (along the Bachmann atrial bundle to the left atrium and along the lower branch of this bundle to the atrioventricular node, where there is a small pulse delay).

Further, the excitation goes to the trunk of the bundle of His, and then to its branch - to the right and left legs. The left leg, in turn, is divided into anterior and posterior branches. In the ventricular myocardium, impulses propagate through Purkinje fibers, which widely anastomose with each other.

Having seized the ventricles with excitement, the impulse that began the path from the sinus node fades because myocardial cells cannot remain excited for a long time. They begin the processes of restoration of their initial state before excitation. These processes are also recorded on the electrocardiogram.

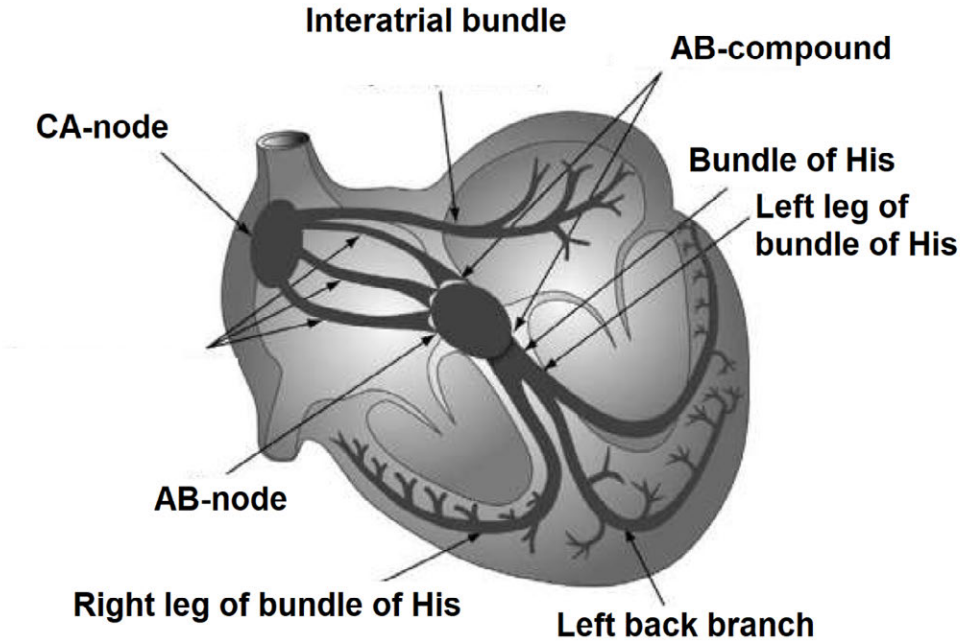

Figure 2.2 The conduction system of the heart [2]

2.2.2 Electrocardiogram (ECG)

The most common device for recording, visualizing and measuring bioelectric potentials of the heart is an electrocardiograph. It consists of an input device, an amplifier of biopotentials and a recording device. An electrocardiograph detects changes in the potential difference between two points in the electric field of the heart, in which the electrodes are installed, during excitation of the myocardium. This registration is called a lead.

Currently, 12 leads are used in clinical practice: three bipolar (standard) leads and nine - unipolar (three reinforced limb leads and 6 chest leads). With bipolar leads, two electrodes are connected to the electrocardiograph, with unipolar leads, one electrode (indifferent) is combined, and the second (trim, active) is placed at the selected point on the body. If the active electrode is placed on a limb, the lead is called unipolar, reinforced from the limb; if this electrode is placed on the chest - unipolar chest assignment.

The passage of an impulse along the conducting system of the heart is graphically recorded vertically in the form of peaks - the rises and falls of the curve line. These peaks are usually called the saw-teeth of the electrocardiogram and denoted by Latin letters [8]. If the saw-tooth amplitude of the QRS-complex from a standard electrocardiograph is more than 5 mm in recording paper, then the tooth is indicated by a capital (capital) letter, if less - a lower case (small) letter.

The time intervals between the same saw-teeth of adjacent cycles are called inter-cycle intervals (for example, P-P and R-R intervals), and between different teeth of the same cycle are called intra-cycle intervals (for example, P-Q and Q-T intervals). The intervals of the electrocardiogram between the waves are designated as **segment**. It's implements in describing properties of a interval – segment, such as duration, displacement towards the isoline or configuration (example of segments in ECG – the ST segment, or RT, the length of the segment from the end of the QRS complex to the end of the T wave). The general view of the electrocardiogram is shown in Figure 2.3.

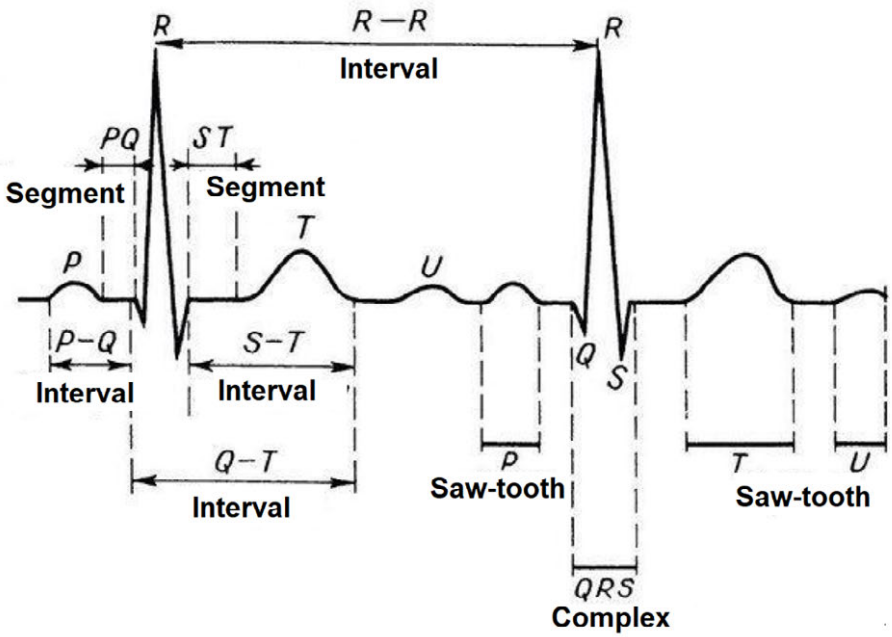

Figure 2.3 General view of the ECG [2]

A typical human ECG consists of five positive and negative defections – saw-teeth, corresponding to the cycle of cardiac activity. They are designated by the Latin letters P, Q, R, S, T, and chest leads (pericardial) - V (V1, V2 V3, V4, V5, V6). Three teeth (P, R, T) are directed up (positive teeth), and two (Q, S) - down (negative teeth).

The amplitude of the teeth is measured in millivolts (mV), 1 mV corresponds to a deviation of 1 cm from the isoelectric line in standard calibrated paper. The width of the teeth and the length of the intervals are measured in seconds. The width of the saw-teeth and the

duration of the intervals are evaluated by the lead, where these parameters have the largest value. Traditionally, all measurements of teeth and intervals are usually made in the second standard lead, denoted by the Roman numeral II.

The P wave reflects the period of atrial excitation, its amplitude varies from 0.05 to 0.25 mV, and the duration is 0.06-0.1 s. The PQ interval corresponds to the time of passage of excitation through the atria and atrioventricular connection to the ventricular myocardium. It varies in duration depending on age, body weight and rhythm frequency. Normally, the PQ interval is 0.12-0.20 s.

The QRS complex is recorded during ventricular arousal, and its duration reflects intraventricular conduction. The amplitude of the saw-teeth of the complex can vary significantly and depends on the position of the heart with respect to the chest leads. The first negative saw-tooth of the complex is called the Q wave, the other negative saw-teeth are called S. The R wave is always positive. The Q wave reflects the depolarization of the interventricular septum. The R wave is the highest in the ECG, it represents the depolarization of the apex of the heart, the posterior and lateral walls of the ventricles. The S wave reflects the excitation coverage of the base of the ventricles.

The ST segment takes the second most important place on the ECG curve. If the appearance of a pathological Q wave indicates necrosis of the heart muscle, then a pathological change in the ST segment may indicate damage to the heart muscle (damage segment). The ST segment reflects the ability (or inability) of the heart muscle at the same time be in state of excitement. In the first case, there is no potential difference and the ST segment is on the isoline or deviates from it by + - 0.1 mV, if there is a lesion, then the potential difference appears and the ST segment deviates from the isoline by a large amount. The T wave reflects the processes of repolarization of the ventricles. In leads from limbs in a healthy person, the amplitude of the T wave does not exceed 0.5-0.6 mV, and in chest leads 1.5-1.7 mV. The duration of the T wave varies from 0.16 to 0.24 seconds.

The R-R interval characterizes the time of one complete cardiocycle, or the time of one heartbeat. In the absence of pathologies in all the leads, standard ECG elements (impulses and intervals between them) are formed, repeating from one cardiocycle to another and reflecting the process of excitation and rest of different parts of the heart muscle. Actually, the ECG is a sequence of cardiocycles repeating at random time intervals. Each individual cardiocycle is defined by a quasi-determined function of complex shape, the sequential components of which have standard letter designations.

The informative parameters of ECG that determine the functional state of the cardiovascular system and are of interest for medical diagnostics are: amplitude, shape, polarity in different leads and repeatability in each cycle of all impulses, duration of P, Q saw-teeth, QRS complex and P-Q time interval, and the PP, QT, RR and PP intervals, as well as deviations from the isoline of the ST segment [3].

In pathology, these parameters can vary significantly, which makes it possible to use electrocardiography in the diagnosis of many heart diseases. Using electrocardiography, various heart rhythm disturbances, coronary insufficiency, various inflammatory and dystrophic lesions, and myocardial infarction are diagnosed [1].

High variability of the parameters of ECG (for example, fluctuations in the amplitude, duration of cardio pulses and intervals between them around average values, local surges and kinks) is observed even in the absence of pathologies and is associated with biological features of the formation and conduct of electrical impulses by body tissues, the location of electrodes and individual properties of a biological objects.

When taking bioelectric signals, the amplification system in the electrocardiograph contributes to a sharp amplification of not only useful signals, but also those minor interference of various types and origin, which are not always eliminated and constitute the main problem in the design and operation of such devices (signal to noise ratios less than 10 dB) In addition, the non-stationarity, complex structure and individual variability of the useful signal should be taken into account.

The presence of interference leads to distortion of the ECG and, accordingly, to the distortion of diagnostic signs. To assess the relationship between useful signal and the interference, Table 1.1 shows the amplitude-time parameters of the ECG corresponding to the norm, and Table 1.2 shows the main statistical parameters of the interference. An examination of the corresponding dependences shows that under favorable removal conditions, compensation for polarization and interference is not particularly difficult, and the interference is mainly represented as a random electromyography process, the spectrum of which has a significant overlap with the ECG spectrum.

The main power of the normal QRS complex is concentrated in the frequency range from 2 to 20 Hz, with a maximum at a frequency of about 15 Hz. In this case, one should take into account a possible change in the spectrum of the ECG depending on the morphology of QRS complexes and heart rate dynamics (with an increase in heart rate, the spectrum shifts toward higher frequencies). Studies show that with pathologies of the functioning of the heart, leading to a change in the shape of QRS complexes (for example, with polytopic ventricular extrasystoles), as well as with physical activity associated with an increase in heart rate, the overlap of signal and interference spectra increases [10].

Table 2.1 - Parameters of the elements of the normal ECG

Parameter	Value parameter of element ECG						
	Saw-tooth P	Interval PQ	QRS-complex	Interval QT	Segment ST	Saw-tooth T	Saw-tooth U
Amplitude, mV	0 - 0.25	-	0.3 - 0.5	-	-	0.4 - 1	0 - 0.1
Duration, s	0.07 - 0.11	0.12-0.2	0.06 - 0.1	0.35 - 0.44	0.06 - 0.15	0.1 - 0.2	-

Table 2.2 - the Parameters of the additive interference accompanying the registration of ECG

Parameter	Value of parameter interference			
	Resting electromyography	Muscle tension electromyography	Crosstalk from the power powerline	Electrode polarization and displacement artifacts
Amplitude, mV	0.01 - 0.05	0.05 - 3	0 - 10^4	0 - 10^3
Duration, s	0 - 300	0 - 10^4	50 and harmonics up to 1000 Hz	0 - 30

The ratio of the frequency properties of the useful signal and interference is presented in Figure 2.4 in the form of graphs of the power spectral density for various components.

Therefore, ECGs are unsteady, complex, structured signals with cyclically repeating informative sections, in the form of bipolar pulses. Based on signs focused on local informative sections of the ECG, the state of the heart is evaluated in electrocardiography. The shape and parameters of informative sections of the ECG are diverse, variable and not always predictable. In addition, interference of various types and origin is inevitably present in the registration channel of the ECG, which is especially pronounced in conditions of prolonged registration and motor activity of patients.

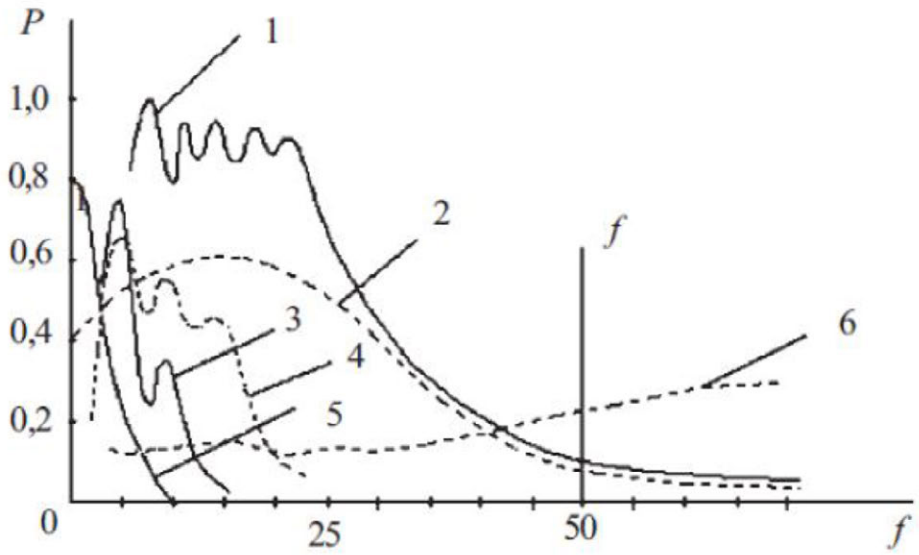

Figure 2.4 - Characteristics of the relative spectral power density of the ECG and noise

The numbers in the figure indicate: 1 - ECG, 2 - QRS complex, 3 - P and T saw-teeth,

4 - artifacts of movement, 5 - drift of the contour, 6 - muscle noise.[9]

A change in the shape of the ECG under the influence of interference leads to a distortion of diagnostic signs, which, in turn, can affect the correct diagnosis. Therefore, to reduce the likelihood of making erroneous or inaccurate diagnostic conclusions, it is necessary to qualitatively process the initial ECG, removing the interference component from the useful component. This leads to the need to develop new methods and means of eliminating interference.

In order to effectively combat interference distorting ECG, first of all, it is necessary to consider their sources, types and characteristics, causes of occurrence, as well as systematize interference according to significant signs. This will be addressed in Chapter 7. In the next two sections we simply provide introduction to ECG signal processing and filtering.

2.3 ECG Signal Processing

Electrocardiographic analysis was one of the very first areas in medicine where computer processing was introduced. Early work mostly dealt with the development of decision tree logic for ECG interpretation, mimicking the rules a cardiologist would apply. It soon became quite evident, however, that the outcome of computer interpretation was critically dependent on the accuracy of the measurements. As a result, the role of signal processing has become increasingly important in producing accurate measurements, especially when

analyzing ECGs recorded under ambulatory or strenuous conditions. In addition, theoretical advances in signal processing have contributed significantly to a new understanding of the ECG signal and, in particular, its dynamic properties.

So far, no system offers a "universal" type of ECG signal analysis, but systems are designed to process signals recorded under particular conditions. It is, therefore, customary to speak of systems for resting ECG interpretation, stress testing, ambulatory ECG monitoring, intensive care monitoring, and so on. Common to all these systems is a set of algorithms which condition the signal with respect to different types of noise, extract basic ECG measurements of wave amplitudes and durations, and compress the data for efficient storage or transmission. The block diagram in Figure 2.5 presents this set of signal processing algorithms, i.e., filtering for noise reduction, QRS detection, wave delineation, and data compression; their respective descriptions define the scope of the present chapter. While these algorithms are frequently implemented to operate in sequential order, information on the occurrence time of a QRS complex, as produced by the QRS detector, is sometimes incorporated into the other algorithms to improve performance. The complexity of each algorithm varies from application to application so that, for example, noise filtering performed in ambulatory monitoring is much more sophisticated than that required in resting ECG analysis.

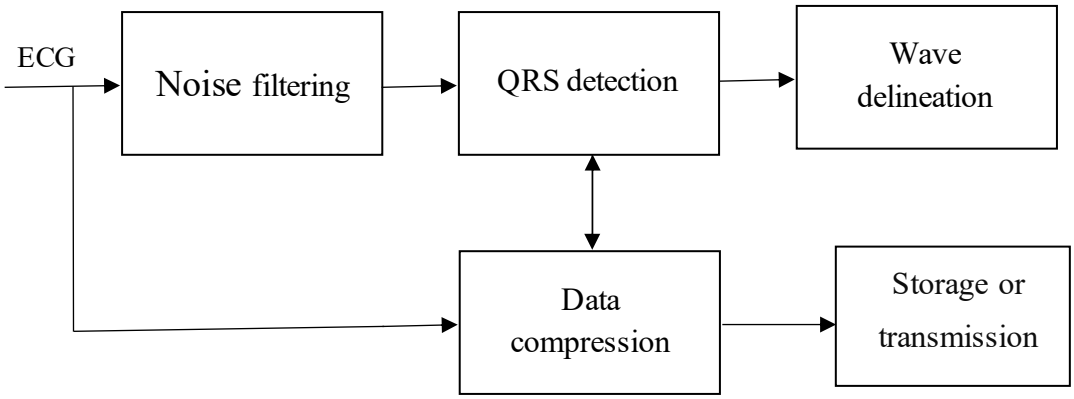

Figure 2.5 Algorithms for basic ECG signal processing.

The timing information produced by the QRS detector may be fed to the blocks for noise filtering and data compression (indicated by gray arrows) to improve their respective performance. The output of the upper branch is the conditioned ECG signal and related temporal information, including the occurrence time of each heartbeat and the onset and end of each wave.

2.4 ECG filtering

Considerable attention has been paid to the design of filters, which may have linear or nonlinear structures, for the removal of baseline wander and other artifacts see Chapter 4,, powerline interference. Removal of noise due to muscle activity represents another important filtering problem being much more difficult to handle because of the substantim spectral overlap between the ECG and muscle noise [10]. This circumstance is identical to the situation

where the EEG signal is disturbed by muscle noise. In contrast to the EEG, muscle noise present in the ECG can be reduced whenever it is appropriate to employ techniques that benefit from the fact that the ECG is a recurrent signal. For example, ensemble averaging techniques used for noise reduction of evoked potentials, described in Chapter 4, can be successfully applied to time-aligned heartbeats for reduction of muscle noise.

The filtering techniques described in Chapter 4 are primarily used for preprocessing of the signal and have as such been implemented in a wide variety of systems for ECG analysis. It should be remembered, however, that filtering of the ECG, as with any other type of biomedical signal, is contextual and should be performed only when the desired information remains undistorted. This important insight may be exemplified by filtering for the removal of powerline interference. Such filtering is suitable in a system for the analysis of heart rate variability, whereas it is inappropriate in a system for the analysis of late potentials, as late potentials spectrally overlap with the interference.

A major concern when filtering out noise is the degree to which the QRS complexes influence the output of the filter. The QRS complex acts, in fact, as an unwanted, large-amplitude impulse input to the filter. Since linear, time-invariant filters are generally more sensitive to the presence of such impulses, filters with a nonlinear structure may be preferable. In order to assure that a filter does not introduce unacceptable distortion, its performance should be assessed by means of simulated signals so that distortion can be exactly quantified.

2.5 Discussion

Electrocardiography - study of heart activity based on registration and analysis of the electrical activity of the myocardium, pervasive heartbeat.

Electrocardiograph consists of an input device, an amplifier of biopotentials and a recording device. An electrocardiograph detects changes in the potential difference between two points in the electric field of the heart, in which the electrodes are installed, during excitation of the myocardium.

Electrocardiogram — is a signal which is displaying the contractile activity of the heart and can be easily recorded using surface electrodes placed on the limbs or chest.

A typical human ECG consists of five positive and negative defections – saw-teeth, corresponding to the cycle of cardiac activity. They are designated by the Latin letters P, Q, R, S, T, and chest leads (pericardial) - V (V1, V2 V3, V4, V5, V6). Three teeth (P, R, T) are directed up (positive teeth), and two (Q, S) - down (negative teeth). In pathology, these parameters can vary significantly, which makes it possible to use electrocardiography in the diagnosis of many heart diseases.

Parameters of ECG reflect:

- QRS complex is recorded during ventricular arousal, and its duration reflects intraventricular conduction.
- P wave – the period of atrial excitation,
- Q wave – the depolarization of the interventricular septum.
- R wave – depolarization of the apex of the heart, the posterior and lateral walls of the ventricles.
- S wave – the excitation coverage of the base of the ventricles.
- ST segment reflects the ability (or inability) of the heart muscle at the same time to be in state of excitement.
- R-R interval – the time of one complete cardiocycle, or the time of one heartbeat.

The informative parameters of the ECG that determine the functional state of the cardiovascular system and are of interest for medical diagnostics are: amplitude, shape, polarity in different leads and repeatability in each cycle of all impulses, duration of P and Q saw-teeth, QRS complex, etc.

Chapter 3

Analysis and systematization of interferences in electrocardiograms

3.1 Introduction

Chapter 3 contains an overview of the main types of interference arising during registration, and the reasons for their occurrence are considered. It describes a suite of methods, essential to any system which performs ECG signal analysis, developed for the purpose of noise reduction, heartbeat detection and delineation, and data compression.

3.1 – introduction.

3.2 – overview of different types of interferences and mistakes in diagnosis caused by presence of interferences in the signal.

3.3 – short overview of the chapter.

3.2 interferences in the ECG.

There are various classifications of interference: by origin and structure, by energy spectrum, by the way of interaction with a useful signal, by probabilistic characteristics and other signs [9].

The interferences arising from the enhancement of biopotentials, according to the method of interaction with the useful input signal (in this case, ECG), are divided into additive and multiplicative. If the interference develops with the signal and their sum acts on the input of the receiver, then this interference is called additive. If the resulting signal is equal to the product of the interference and the transmitted signal, then the interference is called multiplicative.

Additive interference is determined by external influences on the signal transmission medium and does not depend on the values and form of the ECG, and also does not change its informative component.

Multiplicative or deforming interference can change the shape of the information part of the signal, depends on its values and on certain features in the signal, etc. Their value depends on the quality of the communication channel and the quality of service. There are no

standard methods for compensating for multiplicative noise. However, if their nature is known, then a signal correction for the influence of these interference is possible.

Interference arising during registration of the ECG is diverse in origin and structure. This can be external interference, interference from a biological object and interference created by technical means.

Despite the variety of sources, most of the electrocardiographic interference can be combined into only a few groups from the point of view of their manifestation on a registered ECG (and from the point of view of methods for suppressing them). The main obstacles are:

- powerline interferences;
- muscle tremor;
- baseline wander;
- motion artifacts;
- impulse noise;
- high-frequency interferences.

3.2.1 Powerline interferences

The powerline interference looks like a jagged zero line with periodic repetition of the saw-teeths, which correspond to superposition of harmonics of different phases, with frequencies that are multiples of the frequency of voltage fluctuations in the power line. In Europe, the first harmonic with the highest power is 50 Hz. Although a typical form of powerline interference is known in advance, phase is unknown. Moreover, this interference in the general case is not an exact sinusoid, which is manifested by the presence in its spectrum of harmonics from the fundamental frequency.

The reasons for the appearance of powerline interference are poor contact electrodes with skin and improper organization of the research site. Figure 3.1 shows an example of an ECG with powerline interference.

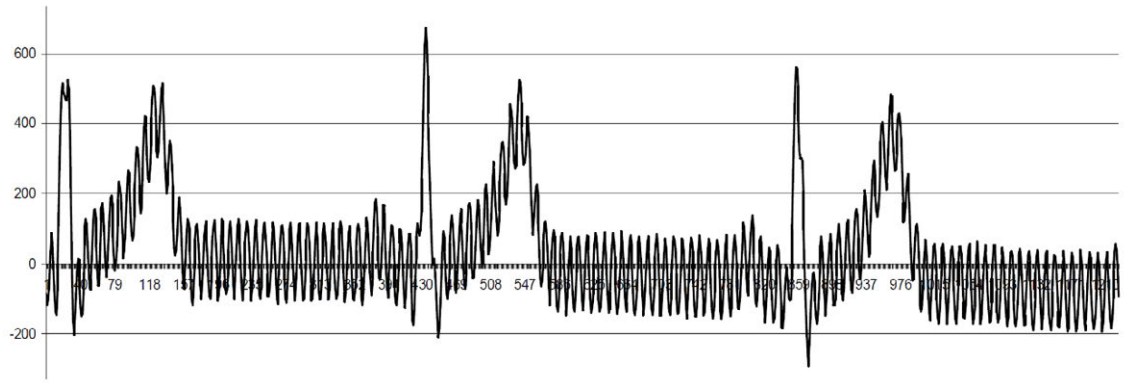

Figure 3.1 - A section of the ECG with powerline interference[2]

3.2.2 Muscle tremor

Muscle tremor looks like a random contour oscillating in a fairly wide range (30 ... 200 Hz). The reason of tremor is the electrical activity of tissues through which the impulse is transmitted (for example, skeletal muscles), the resistance of tissues, especially the skin, as well as the resistance at the input of the amplifier.

Muscle biopotentials are formed by the interference of many random incoherent potentials of individual muscles and are a random sequence of spiky pulses with an amplitude of 0.03…2 mV. The amplitude and frequency parameters of the tremor depends on the physical load and the location of the electrodes. As regards spectral and statistical characteristics of these noises, they are usually close to Gaussian noise with an uneven spectrum, Clifford considers them to be Gaussian flicker noise (the noise power is inversely proportional to the frequency) [10], and Pander models it as unsteady noise, described by a symmetric distribution with "heavy" tails (due to the presence of a pulsed component). Apparently, the latter characteristic is more consistent with reality [11] .

Tremor is superimposed on the ECG of patients with tremulous paralysis, chorea, tetany, parkinsonism. Fluctuations caused by muscle tremors are difficult to distinguish from atrial flutter.

Since skeletal muscles generate a signal independently of the pacemaker, tremor can be considered an additive hindrance. An example of an ECG with muscle tremor is shown in Figure 3.2.

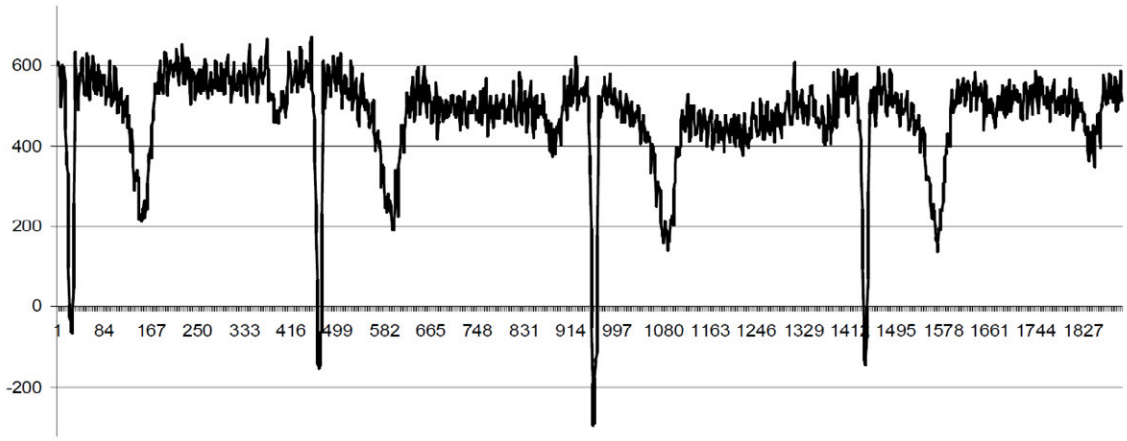

Figure 3.2 – Example of an ECG with muscle tremor [2]

3.2.3 Baseline wander

The baseline wander is a low-frequency oscillation with a typical content below frequency of content below 1 Hz.

Among the most significant reasons for the origin of the baseline wander drift of the ECG, it is worth highlighting [7]:

- motor activity of the patient, in particular respiration and the functioning of the human humoral system;

- change in the electrical properties of the interaction between the patient's skin and the electrode due to disturbed contacts and polarization of the skin under the electrode.

The peculiarity of the baseline wander as a result of physiological influences is the randomness of its behavior and the presence of monotony, often covering several tens of electrocardiocycles.

The contour drift has the greatest influence on the analysis of low-frequency sections of the ECG, especially ST segments. In addition, the variability of the contour affects the accuracy of measuring the amplitude parameters of the teeth, because it is from the isoline that their amplitude is counted. Clifford characterizes this interference with "brown" noise (the noise power is inversely proportional to the square of the frequency) [2]. The stability of the contour depends on the presence of a sufficiently high input impedance of the amplification system and minimal skin resistance. Figure 3.3 shows an example of an ECG with baseline wander.

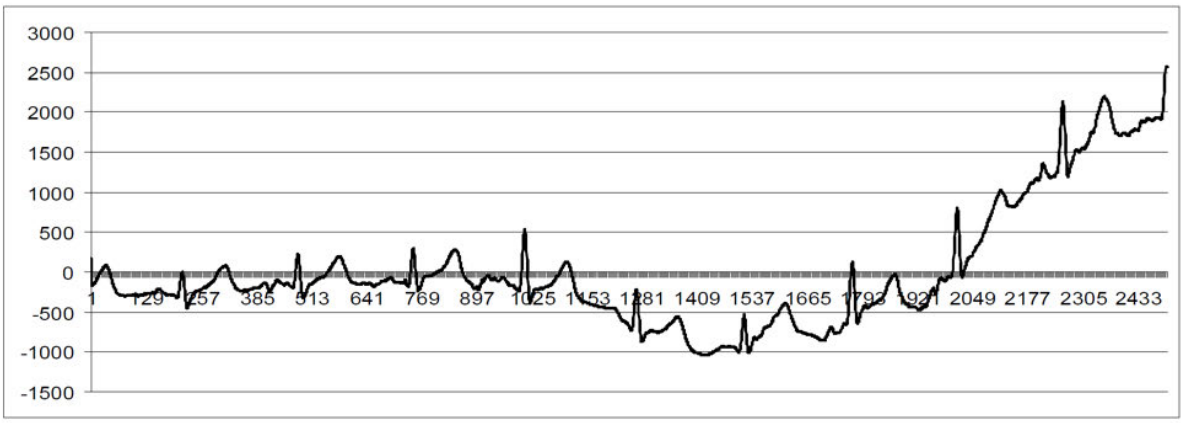

Figure 3.3 - An example of an ECG with baseline wander interference[2]

3.2.4 Motion artifacts

Motion artifacts appear as single or cyclic waves with a frequency in the range of 30-40 Hz. Single artifacts are associated with random mechanical influences on recording equipment, on electrodes in place of contact with skin, cough, hiccups, intestinal motility. Such artifacts in spectrum are very close to the spectrum of QRS complexes and are similar to ventricular extrasystoles, and therefore it is very difficult to differentiate them. Moreover, they pose a significant danger, since they interfere with the detection of QRS complexes, which

can lead to incorrect calculation of heart rate, errors in the analysis of arrhythmias, and incorrect interpretation of the research results. At the peak of the load, motion artifacts can distort the ECG so much that it becomes unsuitable even for visual interpretation.

Cyclic waves are caused by movements of the patient's muscles or small changes in the position of the electrodes during a stress test or during free movement of patients.

3.2.5 Impulse interferences

Impulse interferences is an additive interference that is non-zero only at individual time intervals, separated by significantly longer intervals free from interference. Pulsed interference is a regular or random sequence of interfering pulses. Sources of impulse noise include atmospheric discharges, industrial installations, medical and household electrical appliances, etc [7].

3.2.6 High-frequency interferences

High-frequency noise of electrodes and amplifiers is a random noise of a physical (most often thermal) nature. We can distinguish "Johnson noise", shot noise and flicker noise [15].

Flicker noise is found in all active electronic components as well as some of the passive devices, and like shot noise, is associated with a DC current flow [9]. A characteristic property of this noise is that its magnitude decreases with frequency, and therefore, is sometimes referred to as 1/f noise. The PSD of flicker noise is often modeled by:

$$\tilde{i}_F^2 = K_F \frac{I_{DC}^{\gamma}}{f^{\beta}} \mathrm{A}^2/\mathrm{Hz} \tag{3.1}$$

where K_F,α,and β are constants particular to each device and its fabrication technology and I_{DC} is the DC current through the device.

Flicker noise generally dominates at low frequencies for a properly designed system while the white noise sources become dominant at higher frequencies. Flicker noise corner frequency, f_c is the frequency where the magnitudes of the white and flicker noises of a device are equal. The PSD of a system output is typically similar to the illustration in Figure 3.4.

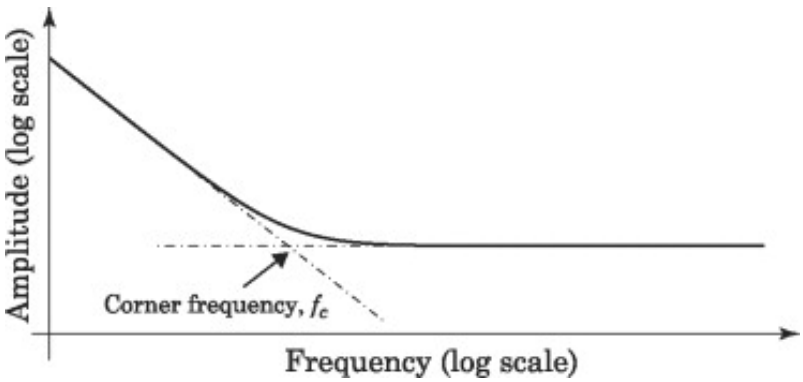

Figure 3.4 -

3.2.7 Errors in diagnosis caused by the presence of interferences in the signal

It should be noted that the spectra of all the above-described interference largely overlap with the ECG spectrum, which leads to difficulties in eliminating them.

According to the spectral characteristics (frequency region in which the main part of interference spectrum is located), all the interference is divided into the following types:

- high-frequency (HF) interference, which includes powerline interference and muscle tremor;
- low-frequency (LF) interference, which includes the contour drift and motion artifact.

In addition, there are other interferences (mainly HF instrumental and electrosurgical noises, some artifacts obtained during registration and processing of ECG, hardware interference, quantization noise and aliasing). However, their spectrum is very different from the spectrum of the ECG, therefore eliminating such interference is already no major problems.

All of the above interferences can cause of errors in measuring the parameters of the ECG and inaccurate detection of signal elements, which, ultimately, can lead to inaccurate or erroneous diagnostic conclusions. On the one hand, overdiagnosis is possible (the system accepts artifacts for various fictitious pathologies), and on the other hand, hypodiagnostics (skipping pathologies) caused by the masking effect of interference. Interference in some electrocardiographic studies (especially those related to patients' locomotor activity) is a serious problem for automatic analysis. The percentage of unsuitable (due to artifacts) to the analysis of the results of Holter monitoring is from 3.7 to 12.4% of studies [9]. The main difficulties in deciphering the results of Holter monitoring are usually associated with artifacts that mimic dangerous cardiac arrhythmias [16]. Even the term "Artifact pseudoarrhythmias" has appeared in the literature. Pseudoarrhythmias can seriously affect the results of the study and, ultimately, the establishment of the correct diagnosis and determination of the entire tactics of patient management.

Another major problem could be not eliminated or poorly eliminated isoline drift (“infarct-like electrocardiogram”), which can be accepted by the automatic analysis system as a false depression or ST segment elevation, and thereby lead to overdiagnosis of myocardial damage [7].

Table 3.1 shows the possible results of erroneous automatic diagnostics in the presence of various types of interference in the ECG. Table 1.4 shows the possible results of

an erroneous diagnosis in the presence of distortions in the ECG. Tables 3.1 and 3.2 were prepared based on an analysis of the literature.

Table 3.1 - the results of erroneous automatic diagnostics in the presence of interference in the ECG

Type of interference	Interim results	Possible erroneous result diagnostics
Artifacts of movements (hiccups, coughing, tremors and Other)	Ventricular extrasystoles, pathological QRS complexes. Atrioventricular condition measurement errors.	Hyperdiagnosis of heart rhythm disturbances
Muscle tremor	Atrial flutter. Atrioventricular condition measurement errors.	Hyperdiagnosis of heart rhythm disturbances.
Baseline wander	Blending or distorting an ST segment. Atrioventricular condition measurement errors.	Myocardial damage. Pericarditis. Myocarditis. Myocardial ischemia. Glycoside intoxication. Violation of repolarization. Hyperdiagnosis of coronary heart disease.
Powerline interference	Errors in detecting QRS complexes. Atrial flutter. Atrioventricular condition measurement errors	Hyperdiagnosis of heart rhythm disturbances.
Single impulse noises	Errors of detection of QRS complexes. Atrioventricular condition measurement errors	Hyperdiagnosis of heart rhythm disturbances.

Table 3.2 - Possible results of erroneous diagnosis in the presence of distortion in the ECG

Type of ECG distortion	Possible result of an erroneous diagnosis
Expansion of the QRS complex	Hypertrophy or dilatation (expansion) of the ventricles. Blockade of any part of the ventricular conduction system.
Reducing the amplitude of the QRS complex	Cardiomyopathy. Diffuse changes in the myocardium due to its defeat in many small focal heart attacks.
Increase in S-wave	Ventricular hypertrophy.
Decrease in the amplitude of the T wave	Damage to the myocardium
Change in the ratio of R and T saw-teeth	Diffuse changes in the myocardium due to its defeat in many small focal heart attacks.
Displacement or distortion of the ST segment	Myocardial damage. Pericarditis. Myocarditis. Myocardial ischemia. Glycoside intoxication. Violation of repolarization.

One of the main requirements for ECG processing algorithms, in addition to suppressing interference, is the preservation of the Q, R, S peaks and smooth extremes of P and T saw-teeth. In case of cardiac abnormalities in the ECG, there may be splits, small peaks, pathological saw-teeth and specific deformations of the QRS complex and ST segment, which are also important to maintain during processing. Therefore, the selection of the ECG filtering algorithm for dynamic properties must be carried out based on the goal of preserving the form of peaks, drops, kinks, smooth extremes, and for statistical characteristics of the relevant signal, while ensuring effective suppression of fluctuations caused by additive and multiplicative noise.

Therefore, it is important to use filters that not only effectively suppress noise, but also do not distort the signal.Accordingly, for the correct diagnosis of cardiovascular diseases and heart diseases, it is necessary to eliminate interference with the obligatory assessment of the quality of their elimination.

Given the wide variety of electrocardiographic interference, evaluating their negative effect on the useful signal and the result of automatic interpretation, requires a more detailed consideration and analysis of methods and the algorithms for suppressing interference in the ECG. In this work we will implement and compare several methods which are widely used for the elimination of baseline wander.

3.3 Discussion

In this chapter we have summarized the electrocardiographic interference:

- **powerline interferences** – superposition of harmonics of different phases, with frequencies that are multiples of the frequency of voltage fluctuations in the power line; looks like a jagged zero line with periodic repetition of the saw-teeths
- **muscle tremor –** looks like a random contour oscillating in a fairly wide range (30 ... 200 Hz). The reason of tremor is the electrical activity of tissues through which the impulse is transmitted (for example, skeletal muscles), the resistance of tissues, especially the skin, as well as the resistance at the input of the amplifier.
- **baseline wander** is a low-frequency oscillation with a frequency of less than 1 Hz, caused by motor activity of the patient and change in the electrical properties of the interaction between the patient's skin and electrode.
- **motion artifacts** appear as single or cyclic waves with a frequency from a few Hz to 30-40 Hz. Single artifacts are associated with random mechanical influences on recording equipment, on electrodes in place of contact with skin, cough, hiccups, intestinal motility. Such artifacts in spectrum are very close to the spectrum of QRS complexes and are similar to ventricular extrasystoles. Caused by movements of the patient's muscles or small changes in the position of the electrodes during a stress test or during free movement of patients.
- **impulse noise** is a regular or random sequence of interfering pulses. Sources of impulse noise include atmospheric discharges, industrial installations, medical and household electrical appliances, etc.
- **High-frequency interferences** includes powerline interference and muscle tremor, which includes motion artifact.

Chapter 4

Baseline Wander removal

4.1 Introduction

Chapter 4 This chapter provides an overview of methods of filtering such as Linear, Time-Invariant Filtering and Forward-backward IIR filtering.

4.1 – introduction in chapter 4.
4.2 – removal of baseline wander summary.
4.3 – Time-Invariant Filtering overview.
4.4 – survey of forward-backward IIR filtering.
4.5 – overview LMS algorithm
4.6 – RLS filtering algorithm
4.7 – short discussion.

4.2. Characteristics of Baseline Wander

Removal of baseline wander is required in order to minimize changes in beat morphology which do not have cardiac origin. This is especially important when subtle changes in the "low-frequency" ST-T segment are analyzed for the diagnosis of ischemia, which may be observed, for example, during the course of a stress test. The frequency content of baseline wander is usually in the range below 0.5 Hz. However, increased movement of the body during the latter stages of a stress test further increases the frequency content of baseline wander. Patients unable to perform a traditional treadmill or ergometer stress test may still be able to perform a stress test by either sitting, running an ergometer by hand, or using a special rowing device. In such cases, baseline wander related to motion of the arms severely distorts the ECG signal. Figure 4.1 shows an example of the ECG with an large amount of baseline wander. The bandwidth of such baseline wander is considerably larger than that caused by respiratory activity and perspiration. We will below describe the two major techniques employed for the removal of baseline wander from the ECG, namely, linear filtering and polynomial fitting. Linear filtering can be further divided into filtering based on time-invariant or time-variant structures.

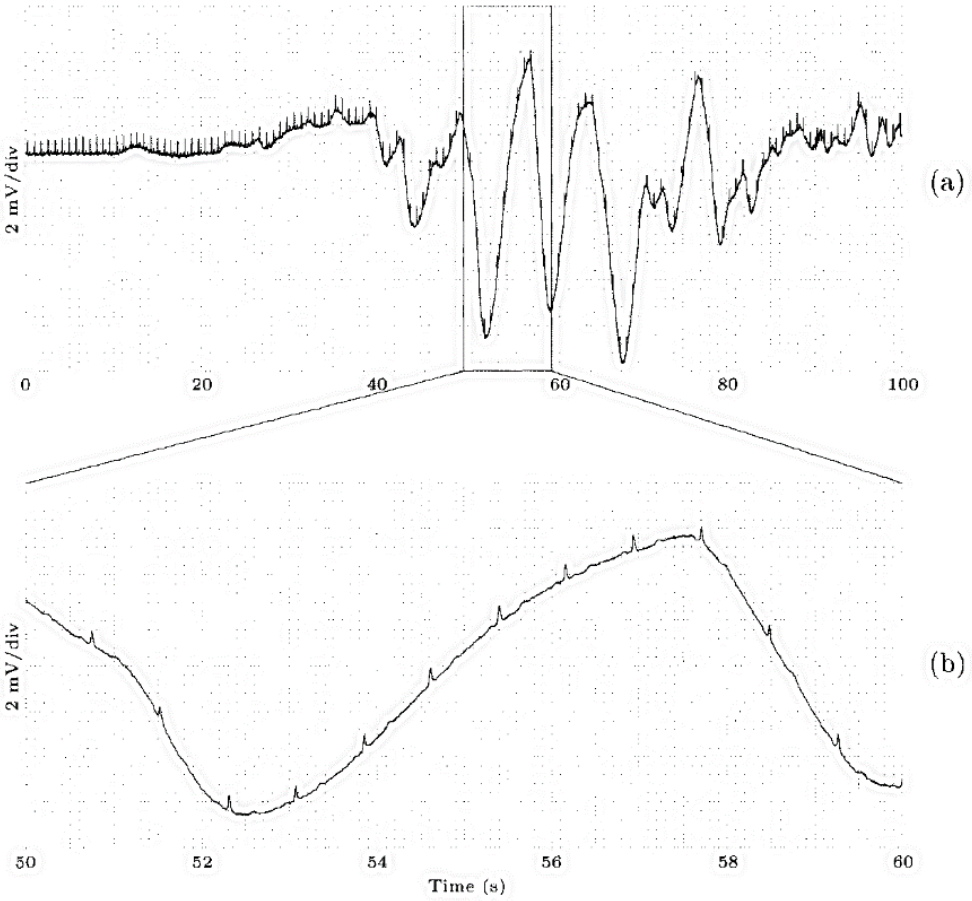

Figure 4.1 - (a) Electrocardiographic baseline wander due to sudden body movements. The amplitude of the baseline wander is considerably larger than that of the QRS complexes. (b) A close-up in time (10x) of the ECG signal framed in (a) [4]

4.3 Linear, Time-Invariant Filtering

The design of a linear, time-invariant, highpass filter to remove baseline wander involves several considerations. The most crucial are the choice of filter cut-off frequency and phase response characteristic. The cut-off frequency should obviously be chosen so that the clinical information in the ECG signal remains undistorted, while as much as possible of the baseline wander is removed [14]. Hence, it is essential to find the lowest frequency component of the ECG spectrum. In general, the slowest heart rate is considered to define this particular frequency component; the PQRST waveform is attributed to higher frequencies. During bradycardia the heart rate may drop to approximately 40 beats/minute, implying that the lowest frequency contained in the ECG is approximately 0.67 Hz. Since the heart rate is not perfectly regular, but always fluctuates from one beat to the next, it is necessary to choose a slightly lower cut-off frequency, approximately Fc = 0.5 Hz. If too high a cut-off frequency is employed,

the output of the highpass filter contains an unwanted, oscillatory component which is strongly correlated to the heart rate [5].

The other crucial design consideration is related to the properties of the phase response and, consequently, the choice of filter structure. Linear phase filtering is highly desirable in order to prevent phase distortion from altering various wave properties of the cardiac cycle, such as the duration of the QRS complex, the ST-T segment level, or the end point of the T wave, see Figure 4.2. It is well-known that FIR filters can have an exact linear phase response, provided that the impulse response is either symmetric or antisymmetric [13, Ch. 4]. On the other hand, IIR filters introduce signal distortion due to the nonlinear phase response.

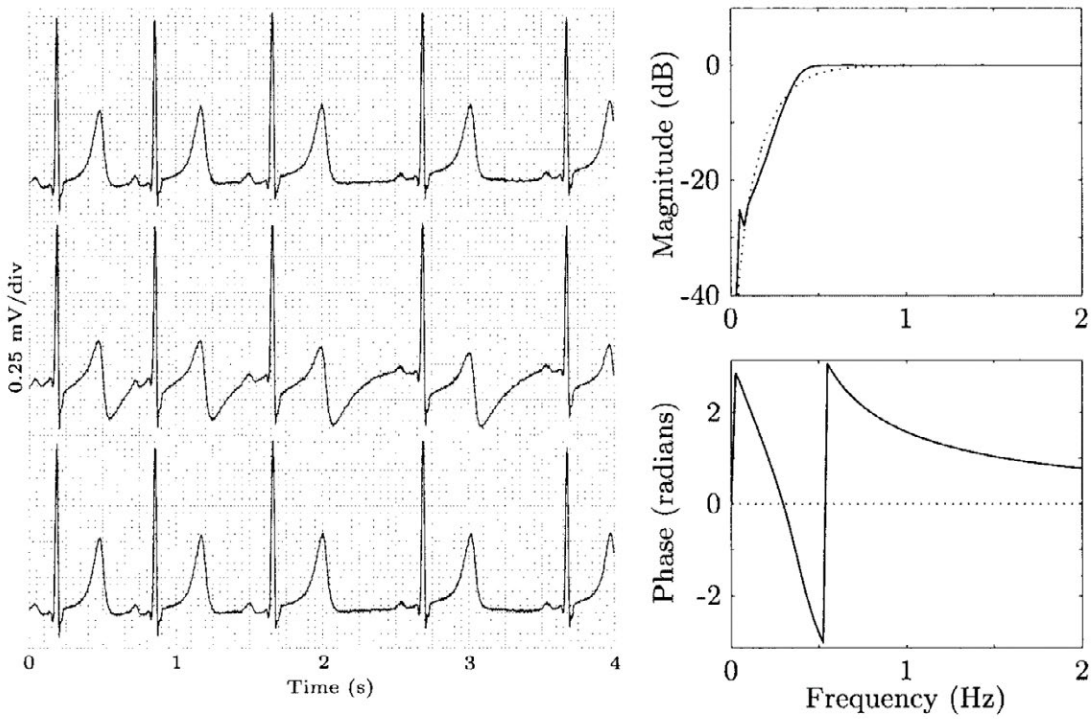

Figure 4.2: Linear, time-invariant, highpass filtering and the effect of a nonlinear phase response. The original ECG (top panel) was processed with two filters having almost identical magnitude functions, but with either a highly nonlinear phase (solid line) or a zero-phase (dotted line). The signal produced by the filter with the nonlinear phase contains severely distorted ST-T segments (middle panel), while the zero-phase filter introduces virtually no distortion at all (bottom panel). The filter cut-off frequency was Fc = 0.5 Hz, and the sampling rate was F8 = 250 Hz [5]

If we assume that the filter cut-off frequency is F~ = 0.5 Hz, and the signal is sampled at a rate of F_s = 250 Hz, the corresponding discretized frequency becomes,

$$f_c = \frac{F_c}{F_s} = 0.002\ (rad) \tag{4.1}$$

which thus establishes that baseline wander removal must be treated as a narrowband filtering problem, i.e., only a fraction of the signal spectrum should be attenuated. Although a linear phase FIR, highpass filter can be designed in numerous ways, the result is invariably a filter with a very long impulse response. A straightforward approach to the design of a filter is to choose the ideal highpass filter as a starting point,

$$H(e^{iw}) = \begin{cases} 0,\ 0 \leq |\omega| \leq \omega_c \\ 1, \omega_c \leq |\omega| \leq \pi \end{cases} \tag{4.2}$$

Since the corresponding impulse response has an infinite length,

$$h(n) = \frac{1}{2\pi}\int_{\omega_c}^{\pi} 1 \cdot e^{i\omega n} d\omega + \frac{1}{2\pi}\int_{-\pi}^{-\omega_c} 1 \cdot e^{i\omega n} d\omega = \begin{cases} 1 - \frac{\omega_c}{\pi},\ n = 0; \\ -\frac{\sin(\omega_c n)}{\pi n},\ n = \pm 1, \pm 2, \dots, \end{cases} \tag{4.3}$$

truncation can be done by multiplying h(n) by a rectangular window function, defined by

$$\omega_n = \begin{cases} 1,\ |n| = 0, \dots, L; \\ 0,\ otherwise \end{cases} \tag{4.4}$$

or by another window function if more appropriate depending on the required attenuation. Such an FIR filter should have an order 2L + 1 is around 1150 at least to achieve a reasonable trade-off between stopband attenuation (at least 20 dB) and the width of the transition band, see Table 7.1. Although the symmetry of the impulse response can be exploited to reduce the number of multiplications in the filter, a considerable number of multiplications is nevertheless required. The use of a filter with lower order, for example, 400, hardly provides any attenuation at all in the stopband, see Figure 4.3.

Table 4.1 - Filter lengths required to achieve a certain stopband attenuation using the window method, in this case with a Keiser window. The cut-off frequency of the highpass filter is 0.5 Hz, the stopband interval is 0-0.3 Hz, and the sampling rate is 250 Hz.

Minimum stopband attenuation (dB)	Required filter length, 2L + 1
20	1142
30	1564
40	1884

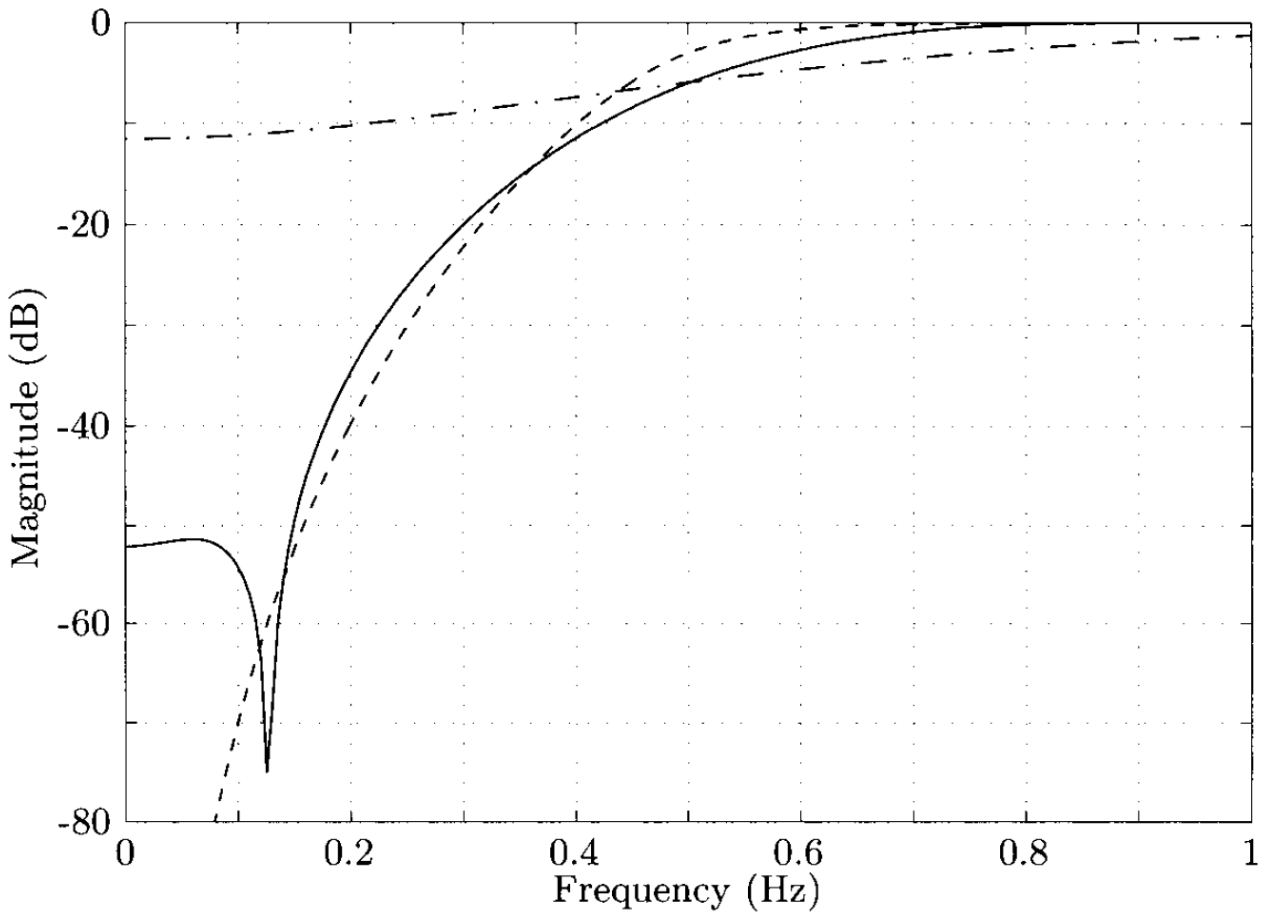

Figure 4.3 - Frequency response for highpass filters designed with the window method, using a Hamming window and a filter length of 400 (dash-dotted line) and 1142 (solid line). The frequency response of the forward-backward, fifth-order Butterworth filter is also displayed (dashed line). The cut-off frequency Fc was 0.5 Hz, and the sampling rate was 250 Hz. [5]

A number of techniques exist with which the above problem of filter complexity can be dramatically reduced, while preserving the linear phase property. These techniques include [2] :

- forward-backward IIR filtering;
- insertion of zeros into an FIR filter;
- sampling rate alteration.

In Section 4.4 we briefly describe the forward-backward IIR filtering technique. For more information on the other two, the interested reader can check [2].

4.4 Forward-backward IIR filtering

An IIR filter meets the magnitude specifications more easily with a much lower filter order due to the freedom of positioning its poles. However, this property is accompanied by a nonlinear phase response. The use of forward-backward filtering remedies this disadvantage since the overall result is filtering with a zero-phase transfer function. Implementation of such a filtering scheme involves three steps: processing of the input signal x(n) with an IIR filter h(n), time reversal of the filter output, and repeated processing with h(n), followed by time reversal of the doubly filtered signal to produce the output signal s(n), or, equivalently,

$$z_1(n) = h(n) * x(n) \tag{4.5}$$

$$z_2(n) = h(n) * z_1(-n) \tag{4.6}$$

$$s(n) = z_2(-n) \tag{4.7}$$

The overall effect of this scheme is established by determining the input output relationship in the frequency domain. Using the discrete-time Fourier transform of a real-valued signal x(n),

$$X(e^{iw}) = \sum_{n=-\infty}^{\infty} x(n)e^{-i\omega n}, \tag{4.8}$$

and the transform property

$$x(-n) \leftrightarrow X^*(e^{iw})$$

(7.5)-(7.7) can be combined and written as

$$S(e^{iw}) = Z_2^*(e^{iw}) = H^*(e^{iw})Z_1(e^{iw}) =$$

$$= H^*(e^{iw})H(e^{iw})X(e^{iw}) =$$

$$= |H(e^{iw})|^2 X(e^{iw}). \tag{4.9}$$

Thus, x(n) is processed with a filter whose magnitude function is $|H(e^{\wedge}(iw))|^2$ and phase function is zero, albeit that h(n) itself has a nonlinear phase response. The order of the overall filter is twice that of h(n). In order to exemplify forward-backward IIR filtering, the overall frequency response for a fifth-order Butterworth filter is shown in Figure 4.4 for fr = 0.5/250. It is evident that the frequency response is close to that of the FIR filter with a length of 1142, but with better attenuation of frequencies close to zero. The number of multiplications required for forward-backward IIR filtering is dramatically lower than that of straightforward FIR filtering. Forward-backward IIR filtering is primarily a scheme for off-line processing since the requirement of causality has to be relaxed when a time-reversed signal is processed. If one is willing to accept a relatively short time delay, this type of filtering can also be implemented in "almost" real time by processing successive, overlapping signal segments. The delay can be

reduced by cleverly choosing the initial conditions of the forward-backward filters such that the initial transients at both ends of the filtered signal are minimized.

The application of forward-backward filtering to baseline wander removal becomes increasingly difficult at higher sampling rates, i.e., for 1000 Hz and higher, since the poles of the filter move closer and closer to the unit circle, or even outside, and the filter thus becomes unstable. Another potential disadvantage of forward-backward filtering is that this technique is not easily extended to handle time-variant filtering in which the cut-off frequency of the highpass filter varies in time.

4.5 Least Mean Squares algorithm

Least mean squares (LMS) algorithms are a class of adaptive filters used to mimic a desired filter by finding the filter's coefficients that relate to producing the least mean squared error signal (difference between the desired and the actual signal). It is a stochastic gradient descent method, in the sense that the filter is only adapted based on the error at the current time.

The (LMS) algorithms adjust the filter's coefficients to minimize the cost function. Compared to recursive least squares (RLS) algorithms, the LMS algorithms do not involve any matrix operations. Therefore, the LMS algorithms require fewer computational resources and memory than the RLS algorithms. The implementation of the LMS algorithms is also less complicated than the RLS algorithms. However, the eigenvalue spread of the input correlation matrix, or the correlation matrix of the input signal, might affect the convergence speed of the resulting adaptive filter [16].

4.5.1 Standard LMS

The standard LMS algorithm performs the following operations to update the coefficients of an adaptive filter:

1) Calculates the output signal $y(n)$ from the adaptive filter.
2) Calculates the error signal $e(n)$ by using the following equation:

$e(n) = d(n) - y(n)$.

3) Updates the filter's coefficients by using the following equation:

$$W(n+1) = W(n) + 2\,\mu(n)\,x(n)\,e(n) \tag{4.10}$$

where μ - is the step size of the adaptive filter

$W(n)$ –is the filter's coefficients vector

$x(n)$ – is the filter's input vector.

4.5.2 Normalized LMS

The normalized LMS (NLMS) algorithm is a modified form of the standard LMS algorithm. The NLMS algorithm updates the coefficients of an adaptive filter by using the following equation:

$$W(n+1) = W(n) + 2\,\mu \frac{x(n)}{\left|\left|x(n)\right|\right|^2} e(n) \tag{4.11}$$

We can also rewrite the above equation as the following equation:

$$W(n+1) = W(n) + 2\,\mu(n)\,x(n)\,e(n) \tag{4.12}$$

In the previous equation, the NLMS algorithm becomes the same as the standard LMS algorithm! except that the NLMS algorithm has a time-varying step size $\mu(n) = \mu/||x(n)||^2$ This adaptive step size can improve the convergence speed of the adaptive filter.

4.5.3 Sign LMS

Some adaptive filter applications require implementing adaptive filter algorithms on hardware targets, such as digital signal processing (DSP) devices, FPGA targets, and application-specific integrated circuits (ASICs). These targets require a simplified version of the standard LMS algorithm. The sign function, as defined by the following equation, can simplify the standard LMS algorithm.

$$\text{sgn}(x) = \begin{cases} 1; & x>0 \\ 0; & x=0 \\ -1; & x<0 \end{cases} \tag{4.13}$$

Applying the sign function to the standard LMS algorithm returns the following three types of sign LMS algorithms.

4.5.3.1 Sign-error LMS algorithm

Sign-error LMS algorithm applies the sign function to the error signal $e(n)$. This algorithm updates the coefficients of an adaptive filter using the following equation:

$$W(n+1) = W(n) + 2\,\mu\,x(n)\,sgn(e(n)) \tag{4.14}$$

Notice that when $x(n)$ is zero, this algorithm does not involve multiplication operations. When $e(n)$ is not zero, this algorithm involves only one multiplication operation: $(2\,\mu)x\,x(n)$

4.5.3.2 Sign-data LMS algorithm

Sign-data LMS algorithm applies the sign function to the input signal vector $x(n)$. This algorithm updates the coefficients of an adaptive filter using the following equation:

$$W(n+1) = W(n) + 2\,\mu\, sgn(x(n))\, e(n) \tag{4.15}$$

Notice that when *x(n)* is zero, this algorithm does not involve multiplication operations. When $x(n)$ is not zero, this algorithm involves only one multiplication operation: (2 *μ)x* $e(n)$.

4.5.3.3 Sign-sign LMS algorithm

Sign-sign LMS algorithm applies the sign function to both *e*(*n*) and $x(n)$. This algorithm updates the coefficients of an adaptive filter using the following equation:

$$W(n+1) = W(n) + 2\,\mu\, sgn(x\,(n))\, sgn(x(n)) \tag{4.16}$$

Notice that when either *e*(*n*) or $x(n)$ is zero, this algorithm does not involve multiplication operations. When neither *e*(*n*) or $x(n)$ is zero, this algorithm involves only one multiplication operations: just adding or subtracting 2 *μ* is required.

The sign LMS algorithms involve fewer multiplication operations than other algorithms. When the step size *μ* equals a power of 2, the sign LMS algorithms can replace the multiplication operations with shift operations. In this situation, these algorithms have only shift and addition operations. Compared to the standard LMS algorithm, the sign LMS algorithm has a slower convergence speed and a greater steady state error [5].

4.6 Recursive least squares filter

Recursive least squares (RLS) is an adaptive filter algorithm that recursively finds the coefficients that minimize a weighted linear least squares cost function relating to the input signals. This approach is in contrast to other algorithms such as the least mean squares (LMS) that aim to reduce the mean square error. In the derivation of the RLS, the input signals are considered deterministic, while for the LMS and similar algorithms they are considered stochastic. Compared to most of its competitors, the RLS exhibits extremely fast convergence. However, this benefit comes at the cost of higher computational complexity [17].

Compared to least mean squares (LMS) algorithms, recursive least squares (RLS) algorithms have a faster convergence speed and do not exhibit the eigenvalue spread problem. However, RLS algorithms involve more complicated mathematical operations and require more computational resources than LMS algorithms.

The standard RLS algorithm performs the following operations to update the coefficients of an adaptive filter:

1) Calculates the output signal *y*(*n*) of the adaptive filter.

2) Calculates the error signal $e(n)$ by using the following equation

$$e(n) = d(n) - y(n) \tag{4.17}$$

3) Updates the filter's coefficients by using the following equation:

$$W(n+1) = W(n) + x(n)\,K(n) \tag{4.18}$$

where $W(n)$ is the filter's coefficients vector and $K(n)$ is the gain vector.

The main goal of RLS filter is to minimize a cost function C by appropriately selecting the filter coefficients w_n, updating the filter as new data arrives. The error signal e(n) and desired signal, d(n), are defined in the negative feedback diagram below (figure 4.4),

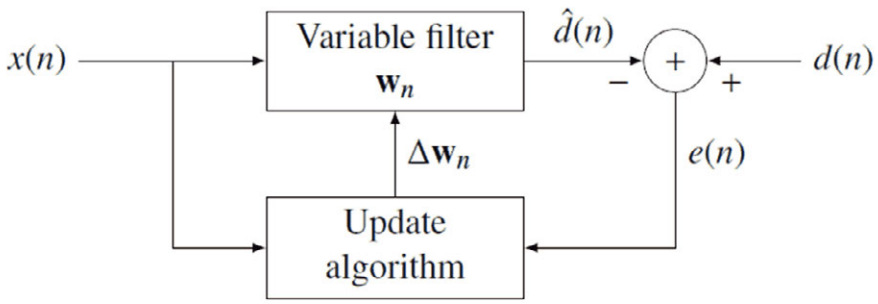

Figure 4.4 - Adaptive RLS filter [19]

To allow for continuous operation, and thus actual implementation, the solution can be computed in recursive form. The output of the FIR filter is computed as

$$y(n) = d(n)w(n-1) \tag{4.19}$$

Since the filter output now incorporates previous samples into the calculation and an a priori estimation error. This is generally different from the a posteriori estimation error, e(n), used in the LMS algorithm.

4.7 Discussion

Baseline wander is a low-frequency artifact in electrocardiogram (ECG) signal recordings of a subject. BW removal is an important step in processing ECG signals, because BW makes interpretation of ECG recordings difficult. The main cause of BW in the ECG signal is movement and respiration of the patient. Baseline wander is usually in the range below 0.5 Hz. Removal of baseline wander is required in order to minimize changes in beat morphology which do not have cardiac origin. This is especially important when subtle changes in the "low-frequency" ST-T segment are analyzed for the diagnosis of ischemia.

Major **techniques for removal of baseline wander** from the ECG:

- **Linear, Time-Invariant Filtering**

The cut-off frequency should obviously be chosen so that the clinical information in the ECG signal remains undistorted while as much as possible of the baseline wander is removed. it is essential to find the lowest frequency component of the ECG spectrum. The slowest heart rate is considered to define this particular frequency component; the PQRST waveform is attributed to higher frequencies. If too high a cut-off frequency is employed, the output of the highpass filter contains an unwanted, oscillatory component which is strongly correlated to the heart rate.

- **Forward-backward IIR filtering**

IIR filter meets the magnitude specifications more easily with a much lower filter order due to the freedom of positioning its poles. This property is accompanied by a nonlinear phase response.

The linear phase implementation involves three steps: processing of the input signal x(n) with an IIR filter h(n), time reversal of the filter output, and repeated processing with h(n), followed by time reversal of the doubly filtered signal to produce the output signal s(n). The overall effect of this scheme is established by determining the input output relationship in the frequency domain.

Forward-backward IIR filtering is primarily a scheme for off-line processing, since the requirement of causality has to be relaxed when a time-reversed signal is processed. If one is willing to accept a relatively short time delay, this type of filtering can also be implemented in "almost" real time by processing successive, overlapping signal segments. The delay can be reduced by cleverly choosing the initial conditions of the forward-backward filters, such that the initial transients at both ends of the filtered signal are minimized.

LMS algorithm has established itself as the workhorse of adaptive signal processing for two primary reasons: computational efficiency and robust performance. However, for signals with a large spectral dynamic range, the LMS has a non-smooth and slow rate of convergence. In addition, if the signal is also non-stationary (e.g., speech and audio signals), then the LMS can be an unsuitable adaptation method. LMS algorithms are known as slowly converging algorithms. The speed of convergences define the number of signal intervals that are necessary to obtaining reliable filter coefficients. [16]

RLS algorithm have much better starting convergence properties but is more complex and for that reason, not so suitable for low computational devices. RLS method, has better convergence rate and less sensitivity to the eigenvalue spread than LMS. [17]

Chapter 5

Results

5.1 Introduction

This chapter presents the results for the validation process of these techniques such as Linear Time-Invariant Filtering and Forward-backward IIR filtering.

5.1 – introduction.
5.2 – removal of baseline wander.
5.3 – short discussion.

5.2. Performance of denoising algorithms

During the implementation process, all the validation tests were performed in a x86 platform without any constraint in terms of memory consumption and computation cost. The implementation was validated using the Physionet MIT-BIH Arrhythmia Database (mitdb) [15].

Noisy ECG signals were created by adding artifacts to clean the ECG signals. The clean ECG signal was 1 minute duration of the signal record 100 from the MIT-BIH database. All artifacts were also obtained from the MIT-BIH database. The noisy ECG signals is illustrated in Figure 5.1. Once the noisy ECG signals were created through adding the baseline wander interference to clean signal, they were fed into main MATLAB program. In the main program, performance of LMS, NLMS, KLMS, NKLMS and RLS algorithms on the noisy ECG signals were tested. For achieving the best result in filtering in experimental way were manually selected parameters for input, such as stepsize in range from 0.1 to 250 and mu in range from 1e-10 to 1, respectively. This process was done for each LMS, NLMS, KLMS and NKLMS algorithms, through changing methods to calculate filter weights. Method to calculate filter weights, specified the following:

'LMS' – Solves the Weiner-Hopf equation and finds the filter coefficients for an adaptive filter.

'Normalized LMS' – Normalized variation of the LMS algorithm.

'Sign-Data LMS' – Correction to the filter weights at each iteration depends on the sign of the input x.

'Sign-Error LMS' – Correction applied to the current filter weights for each successive iteration depends on the sign of the error, err.

'Sign-Sign LMS' – Correction applied to the current filter weights for each successive iteration depends on both the sign of x and the sign of err.

Choosing optimal parameters for designed filter is important. In ECG filtering this parameters are calculated through approximation, manually best performance by experimenting. For baseline wander removal – the smaller the step-size is, the better filter will perform. As a disadvantage, in this case filter will converge, but filtered signal will be deprived of more artifacts.

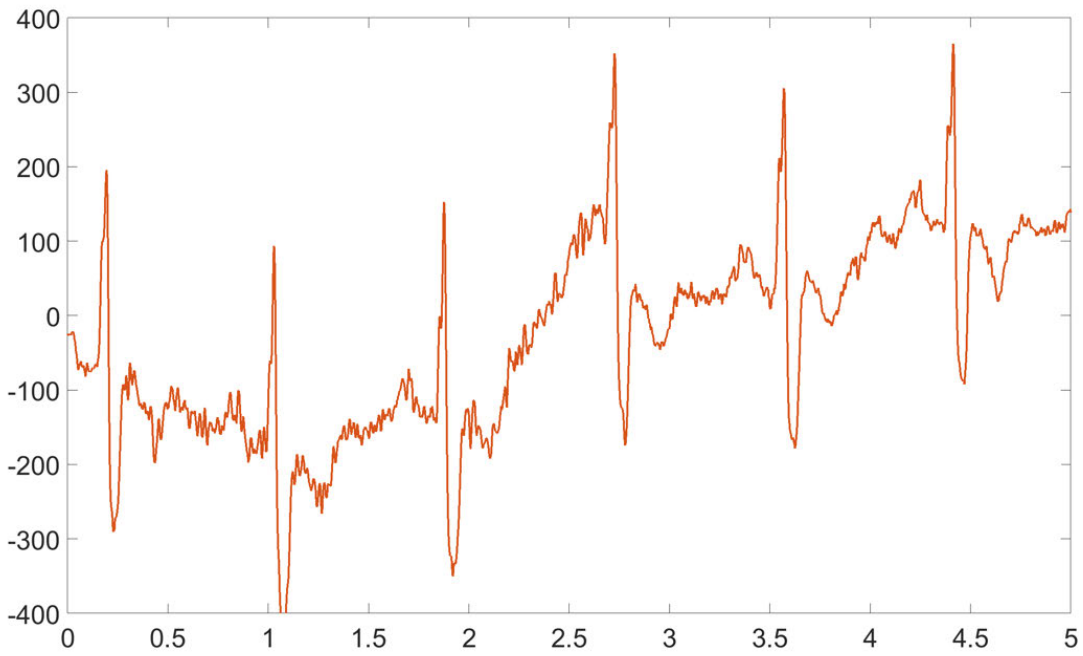

Figure 5.1 – ECG signal with baseline wander interference

The **mean squared error (MSE)** of an estimator (a procedure for estimating an unobserved quantity) measures the average of the squares of the errors—that is, the average squared difference between the estimated values and the actual values. MSE is a risk function, corresponding to the expected value of the squared error loss. That MSE is almost always strictly positive can be attributed to either randomness, or the fact that the estimator does not account for information that could produce a more accurate estimate [7].

The speed with which the filter converges to the optimal state, known as the convergence speed, depends on multiple factors such nature of the input signal, choice of the adaptive filter algorithm, and step size of the algorithm. Inputs in LMS filter which were used: mu and stepsize.

Setting the step size of ***μ*(mu)** needed for convergence of the normalized LMS equations $0<\mu<2$.

Stepsize – adaptation step size factor, specified as a non-negative scalar. The step size increases or decreases as the mean-square error increases or decreases, allowing the adaptive filter to track changes in the system as well as produce a small steady state error.

Small step size ensures a small steady state error between the output y and the desired signal d. If the step size is small, the convergence speed of the filter decreases. To improve the convergence speed, increase the step size. Note that if the step size is large, the filter can become unstable:

- too large step size gives a fast response to plant changes but results in a large excess mean square error (MSE), and may even cause loss of convergence;

- too small step size degrades tracking capabilities of the algorithm.

An optimal step size, giving a trade-off between the speed of convergence and residual error, depends on the power of the input data.

Therefore, the main, commonly used modification of the LMS algorithm is a normalization of the step size. This leads to the normalized LMS (NLMS) algorithm, in which the step size is scaled (divided) by an estimated power of the input data. However, even the NLMS algorithm requires a "base" step size choice[7].

We have applied denoising methods overviewed in Chapter 4 on ECG signal with artifacts. The parameters for implemented filters and the normalized MSE(NMSE) are shown in Table 5.1.

Table 5.1 - Summary of algorithm differences

Denoising method	Parameters			NMSE before applying filters
	mu	Stepsize	NMSE	
Highpass filter	-	-	0.1698	1.1847
Standard LMS	0.0000000005	200	0.1929	
Normalized LMS	0.0191	216	0.2083	
Sign-Data LMS	0.00000074	202	0.2083	
Sign-Error LMS	0.0000001	200	0.2120	
Sign-Sign LMS	0.0000195	199	0.2201	
RLS	-	11	0.0758	

Figure 5.2 showing the results of modeling an ECG signal after applying highpass filter.

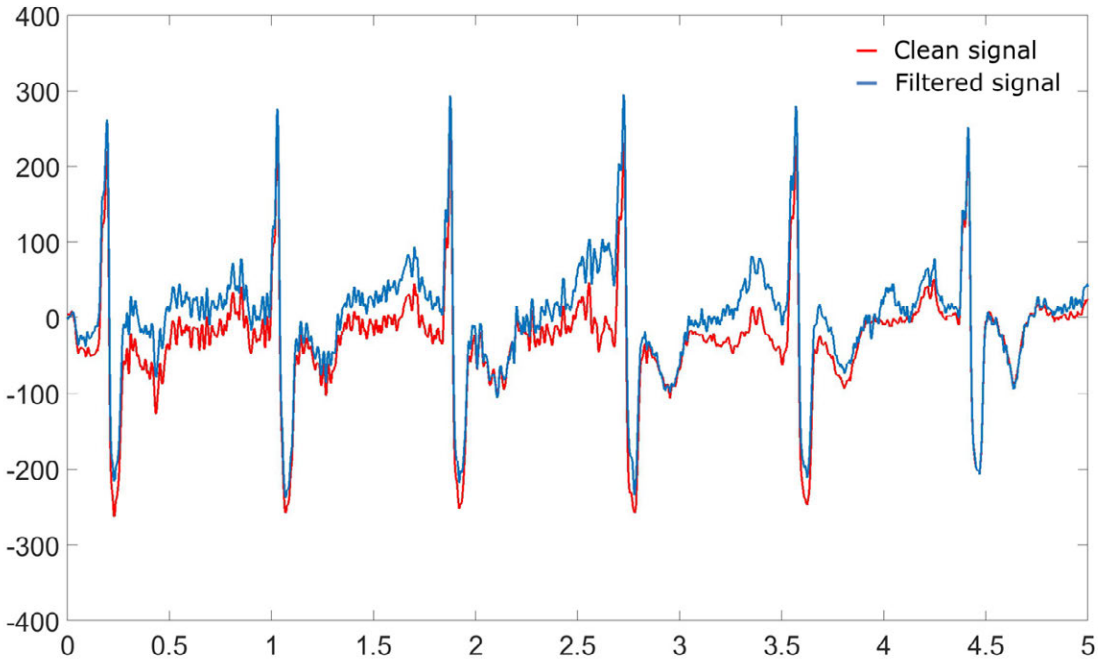

Figure 5.2 – Filtered ECG signal with highpass filter.

The signals shown in Figures 5.3 and 5.4 correspond to the outputs the adaptive filters. The adaptive filters used are the LMS and RLS algorithms respectively. We have also considered other types of adaptive filters: NLMS, SDLMS, SELMS, SSLMS. But they showed similar results a LMS filter, shown in Figure 5.3. This result also could be overviewed in Table 5.1.

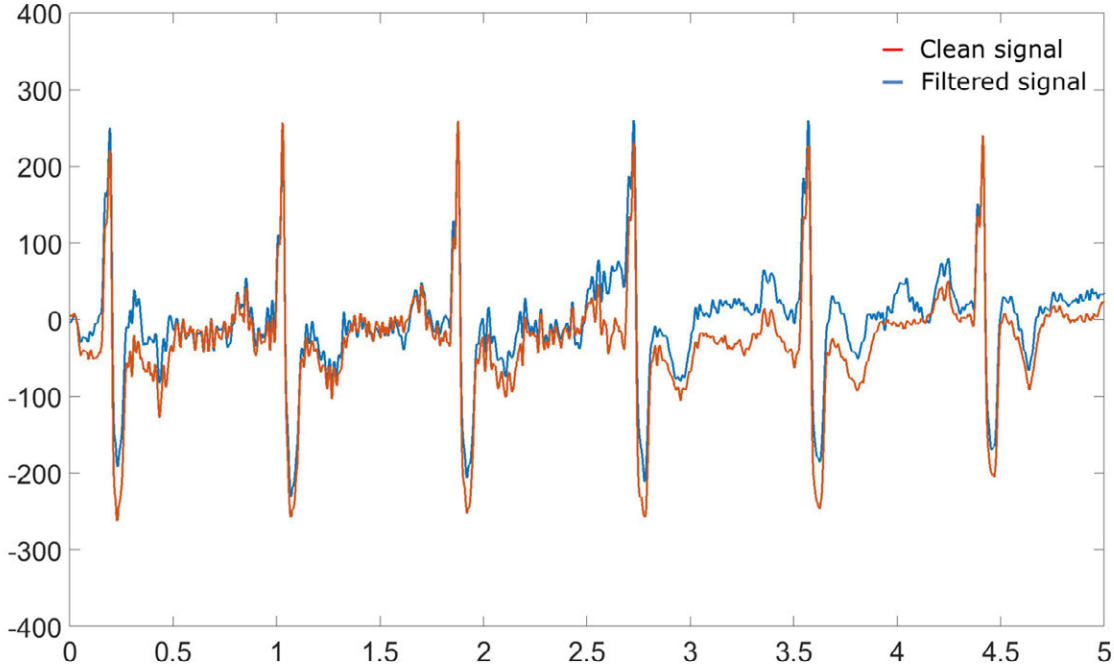

Figure 5.3 – Filtered ECG signal with LMS filter

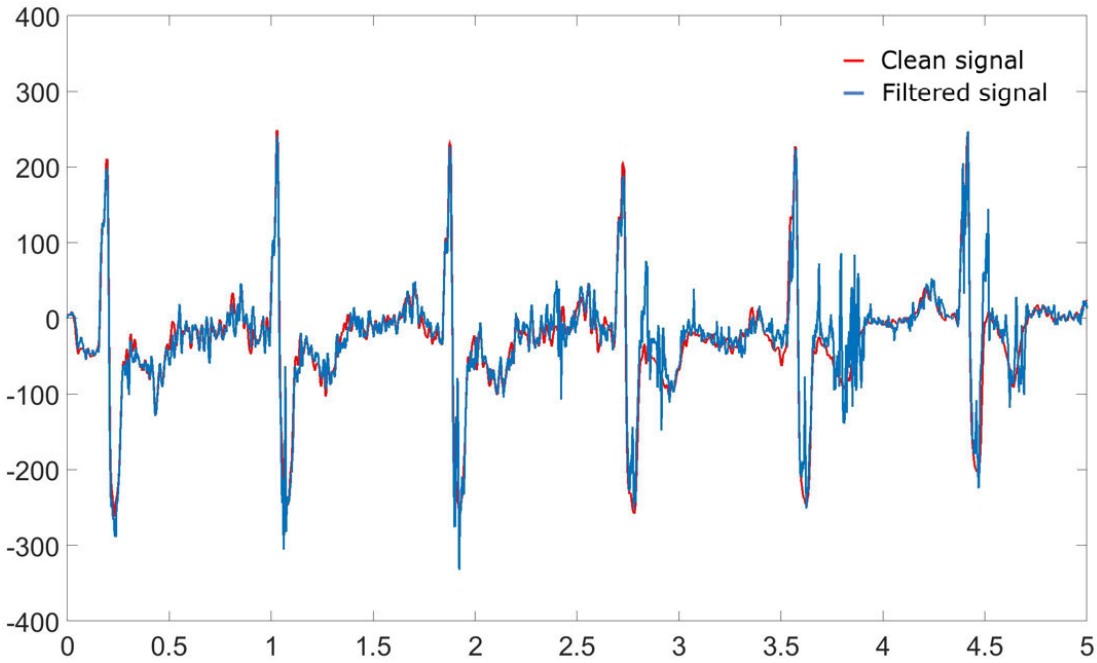

Figure 5.4 – Filtered ECG signal with RLS filter

5.3 Discussion

Based on results it can be seen that the RLS algorithm showed best result in filtering. Even that after RLS filtering signal received other type of noise, all peaks of the ECG are not distorted, it means that main information from ECG signal can be used in diagnosing. However, even though this program is designed for online computation, it still takes a while to compute, as discussed in Table 5.2. As for improving the results, one can implement a variable step-size algorithm which can help on faster convergence of algorithm at the start.

Table 5.2 - Summary of LMS and RLS algorithms

Algorithm	Data modeling	Computation	Simplicity
LMS	Linear	Fast	Simple
Normalized LMS	Linear	Fast	Simple
Sign LMS	Linear	Fast	Moderate
RLS	Linear and non-linear	Moderate	Moderate

Conclusions and future work

The examination of the ECG has been comprehensively used for diagnosing heart diseases. Various techniques have been proposed earlier in the literature for reduction of baseline wander interference from ECG. This paper provides an overview of various filtration techniques available in the literature for removal of Baseline Wander interference. Literature indicates that the filtration techniques for ECG must be highly accurate and should ensure fast filtration. In the present paper effort has been made to perform the comparative analysis of different filters that were proposed earlier by various authors for suppression of base line wander interference.

The ECG signal is corrupted by different types of noises as a high frequency noise and low frequency noise. The algorithms which are used in this project can improve the performance of filtering for these types of signals. As a conclusion, baseline wander removal is a filtering need which can be implemented efficiently in an embedded platform. In this project, we have developed several implementations for baseline wander removal and the results of posterior delineation show that it helps to process the ECG signal in noisy contexts in an embedded platform.

In this project FIR and IIR filters were used to remove the low frequency noise. The output of the FIR filter is very near about the desired value and has small removal of low frequency noise while the output which we get through IIR filter has less value than the desired output. Another important conclusion which we get through the observation is this that the standard deviation of IIR filter is much closer to the desired output & the standard deviation of the FIR filter output with high frequency noise have less value than the IIR filter. The results which are given by FIR filter for proposed algorithm have much accurate results compared with IIR filter.

For future work, it would be needed to use another source of ECG signal instead of the Physionet database. We have validated the implementations by using a program to send the node the input samples whereas this project stands for realtime ECG processing in a wearable platform. It has been proved that implementations here presented are acceptable to run in an embedded platform, so the next step may have been to perform a real ECG ambulatory processing by the use of a node. Also future work may concentrate on designing of filters for accurate and fast filtration of ECG which ultimately results in the improvement of accuracy during diagnosing the cardiac disease at the earliest in the use of patient monitoring systems.

Regarding the implementation of a Wireless Body Sensor Network, these implementations would be only the step in a more complex system but, at the same time, a filtering stage is needed for posterior processing. Because of the optimizations made, the morphological filtering implementation offers a good performance to support posterior operations.

According to the results of noise suppression is needed. As exposed in the first chapter, noise suppression is a major challenge in ECG processing and there is not a suitable method yet. Approach to noise suppression may result in a better performance.

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